Linear-probe diagnostics of observed and source-injected strong-lens images in DESI Legacy Imaging Surveys DR10

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ABSTRACT

Injection–recovery characterizes CNN strong-lens finders, but its interpretation depends on the simulated images presented to the detector. For a frozen EfficientNetV2-S trained on DESI Legacy Imaging Surveys DR10 cutouts, we measure conditional recovery and observed–injected distinguishability separately. Matched and paired controls identify differences already present before source insertion and measure detector responses to specified construction changes. Dependency-aware linear probes compare four representations—the operational detector's pooled penultimate features, self-supervised survey features, ImageNet features, and pixel statistics—for 43 selected observed systems and 6,336 source-injected images. The populations separate in all four representations, with out-of-fold receiver-operating-characteristic AUCs of 0.991, 0.958, 0.979, and 0.981, respectively. A leading pixel-statistic offset is already present between observed systems and bare targets. Along a separately fitted operational observed–bare axis, the mean injection movement is near zero despite heterogeneous pairwise effects. Separately, the finder recovers 3.45 per cent of a 49,500-scene grid with no visibility cut. Modelled source-photon sampling lowers recovery, whereas the single-Sérsic construction is recovered more often than its paired structured-source counterpart. On 43 fixed targets, expectation-image recovery varies non-monotonically with source brightness and peaks at 18.6 per cent. Together, these analyses reveal differences between the selected populations within the actual finder, identify a pre-insertion offset, and measure responses to controlled injection choices. They provide a practical basis for comparing future injection pipelines using trained finders.

Key words: gravitational lensing: strong – surveys – techniques: image processing – Data Methods – Machine Learning

1 INTRODUCTION

Strong gravitational lenses constrain galaxy mass profiles, dark-matter structure, and cosmological distances (e.g. Treu 2010; Collett 2015). Building large samples requires scalable discovery methods, including convolutional neural networks (CNNs). Population inference also requires a selection function: a model of whether a lens is physically detectable and whether a specified search procedure finds it (Sonnenfeld et al. 2023; Sonnenfeld 2024). Injection–recovery measures the response after synthetic lensed light is added to survey images. Without a separate detectability condition, its recovery fraction combines physical visibility with finder response.

CNN lens finders perform well on simulated benchmarks and labelled observational samples (e.g. Petrillo et al. 2017; Lanusse et al. 2018; Jacobs et al. 2019; Cañameras et al. 2021; Storfer et al. 2024), but their calibration depends on the simulations used. Some pipelines inject parametric Sérsic sources lensed by idealised mass models (e.g. Herle et al. 2024); others use images of observed galaxies as source-plane templates (e.g. Cañameras et al. 2021; Rojas et al. 2022; Schuldt et al. 2023). Cross-pipeline results combine differences in target selection, lens models, source distributions, image processing, and source structure (Jaelani et al. 2024; Cañameras et al. 2024). A classifier two-sample test asks whether a model can tell which of two samples an object came from using only a chosen representation (Lopez-Paz & Oquab 2017). In strong-lens work, self-supervised survey representations have supported lens discovery (Stein et al. 2022), while domain-adaptation methods have explicitly addressed differences between simulated and observed lens data (Parul et al. 2025). A linear probe is an L_2-regularised logistic regression fitted to frozen features; its receiver-operating-characteristic area under the curve (AUC) measures how well the two samples are separated on held-out data.

Injection recovery measures how often a finder responds to a specified simulated population. A complementary representation-level comparison asks how clearly observed and injected populations separate in a chosen feature space. We therefore use a linear probe of the operational detector's pooled penultimate features as a detector-specific diagnostic. It reuses the trained representation to evaluate a held-out observed–injected readout for the fixed finder. We apply the same dependency-aware procedure to self-supervised survey features, generic ImageNet features, and elementary pixel statistics to map how broadly separation appears across representations. Comparing the four protocols shows whether separation is specific to the lens-trained network or also appears in unrelated learned and pixel-level representations.

We interpret the comparison using the existing lens-classification score, native-pixel summaries, and matched observed–bare–injected controls. We also measure how the fixed detector responds to the injection grid, source-photon sampling, source construction, and source brightness. Together, these analyses distinguish population-

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level representation differences from responses to controlled construction choices.

2 DATA, SAMPLES, AND DETECTOR

We use DR10 cutouts to measure injection recovery and feature-space differences between the observed and simulated samples.

2.1 DESI Legacy Imaging Surveys DR10

We use g-, r-, and z-band coadded imaging from the tenth data release (DR10) of the DESI Legacy Imaging Surveys (Dey et al. 2019).[1] The images have a pixel scale of 0.262 arcsec pixel^{-1}, and the nominal 5σ point-source depths of the Legacy Surveys are $g \approx 24.7$, $r \approx 23.9$, and $z \approx 23.0$ mag (AB). Because the delivered point-spread function (PSF) and depth vary across the footprint and affect the detectability of lensed features, these nominal depths characterise the survey design rather than every location in DR10. Where these quantities enter the injection experiments, we therefore use the local catalogue measurements at each injection-target position rather than assigning one survey-wide value.

The Tractor catalogue is the model-based source catalogue produced from the Legacy Surveys imaging (Lang, Hogg & Mykytyn 2016; Dey et al. 2019).[2] For each catalogue object used in this study, we extract a 101×101 pixel cutout (26.5×26.5 arcsec^2) centred on its Tractor position. Cutouts are stored in nanomaggies, for which the AB zeropoint is 22.5, and the g, r, and z images form the three detector input channels; no other DR10 catalogue or imaging products are used as detector inputs in this analysis.

2.2 Preprocessing

We apply the same preprocessing to observed and injected cutouts. For each band b, let A denote the annulus containing pixels whose radial distance from the cutout centre satisfies $20 \leq r < 32$ pixels. We define

$$m_b = \mathrm{median}_{i \in A}(x_{b,i}), \qquad s_b = \mathrm{median}_{i \in A}|x_{b,i} - m_b| + 10^{-8}, \quad (1)$$

and transform each pixel as

$$x'_{b,i} = \mathrm{clip}\left(\frac{x_{b,i} - m_b}{s_b}, -10, +10\right). \quad (2)$$

Thus, each band is centred and scaled using a robust local reference while preserving the full 101×101 field. The scale s_b is the unscaled median absolute deviation (MAD), rather than a Gaussian-equivalent standard deviation.

All detector comparisons retain the 20–32 pixel normalisation annulus used during training. The annulus can contain light from the central object or injected source, so its median and MAD can depend on astrophysical structure as well as the local background. These statistics define the detector's trained input mapping.

[1] DR10 description: https://www.legacysurvey.org/dr10/description/, accessed 29 August 2026.
[2] DR10 catalogue documentation: https://www.legacysurvey.org/dr10/catalogs/, accessed 29 August 2026.

2.3 Detector development and evaluation roles

Table 1 summarizes the samples used for training, validation, threshold calibration, and evaluation. Development positives are noisy labels from the literature-reviewed candidate set and the lower-confidence catalogue-candidate set; confirmed lenses are reserved for evaluation. False-positive rate (FPR) denotes the fraction of catalogue-negative examples at or above threshold. We calibrate the threshold on one negative sample and estimate its FPR on another.

2.4 Labelled samples and provenance

Lenscat v1.1.3 is the public, community-contributed strong-lens catalogue compiled by Vujeva et al. (2025) and archived by Lo & Vujeva (2025).

The positive parent catalogue was assembled from Lenscat. We selected 5,104 entries whose names identified them with DESI imaging; 4,788 were associated with a Tractor source in the processed DR10 sweeps within 5 arcsec. Lenscat places 389 of these systems in its highest-confidence category and 4,399 in a lower-confidence candidate category. All 389 highest-confidence records underwent object-level literature review. The 4,399 lower-confidence records supplied noisy positive labels for detector development, subject to the role exclusions in Section 2.6. Lenscat confidence was not treated as confirmation because its highest category can denote either spectroscopically confirmed multiple images or the highest grade assigned by a contributing study (Vujeva et al. 2025). The entries originate from heterogeneous discovery samples, including the DESI Legacy Imaging Surveys searches of Huang et al. (2020, 2021), Stein et al. (2022), and Storfer et al. (2024).

Literature review identified 63 confirmed lenses among the 389 reviewed systems. Of these, 50 have secure spectroscopic redshifts for both deflector and source, seven have secure lens evidence but incomplete or lower-confidence spectroscopy, and six are confirmed by imaging or lens modelling without secure redshifts for both components. The available literature provides heterogeneous spectroscopic and imaging support across the cohort. The four non-lenses and two uncertain systems are excluded from lens-performance estimates; the remaining 320 candidates form the reviewed source of noisy positive labels for detector development. For the 43-system evaluation cohort, object-level confirmation evidence and available deflector and source redshifts are listed in Appendix Table A11.

The negative parent sample was drawn from the Tractor catalogue in two stages. The sampling frame required finite coordinates, $-18^\circ < \delta < +32^\circ$, a Tractor morphological type of SER, DEV, or REX, at least three exposures in each of g, r, z, and none of mask bits 1, 5, 6, 7, 11, 12, or 13. It also required a separation of at least 11 arcsec from the 5,104-entry DESI-named Lenscat parent catalogue. For rows with positive fitted z-band flux, the observed magnitude, not corrected for Galactic foreground extinction, had to satisfy $z < 20$. Rows with finite non-positive z-band flux have no logarithmic magnitude and were therefore retained without the $z < 20$ requirement; the available records do not document a scientific reason for this exemption. Colour was not an eligibility cut, but $g-r$ contributed to the sampling strata described below. All reported analyses use this fixed sample. The available sampled-row table retains the z-band exposure count but omits the corresponding g- and r columns. The archived roster and image cutouts are sufficient to reproduce the reported analyses, but the available table does not contain the g- and r-band exposure counts needed to recheck those two eligibility criteria row by row.

Rows in the sampling frame were assigned to two sampling strata.

Table 1. Detector-development and evaluation roles. "Reviewed candidates" are Lenscat systems classified as candidates—rather than confirmed lenses, non-lenses, or uncertain systems—after object-level review of the published evidence. "Additional catalogue candidates" are lower-confidence catalogue records used only as noisy positive examples. The calibration, achieved-FPR, and selected-system samples are disjoint and are used only after training.

Role	Negative examples	Positive or observed examples	Function
Training: reviewed + additional	286 679	238 reviewed candidates + 3 395 additional catalogue candidates	Primary training set; 290 312 rows
Validation: reviewed + additional	61 437	64 reviewed candidates + 714 additional catalogue candidates	Development diagnostics and sensitivity-checkpoint selection; 62 215 rows
Training: reviewed candidates only	286 679	238 reviewed candidates	Alternative positive-label composition; 286 917 rows
Validation: reviewed candidates only	61 437	64 reviewed candidates	Checkpoint selection for the corresponding runs; 61 501 rows
Full test partition	61 435	63 confirmed systems	Source partition for final evaluation roles
Threshold calibration	20 002	None	Set the primary or checkpoint-specific operating threshold
Achieved-FPR evaluation	12 512	None	Evaluate catalogue-negative threshold exceedance
Selected-system evaluation	None	43 confirmed systems	Recovery among the 43 selected confirmed systems
Exploratory-analysis-development role	None	20 confirmed systems	Excluded from endpoint and observed-reference denominators

Table 2. Literature-based classifications for the 389 reviewed Lenscat records.

Scientific status	Number
Confirmed lens	63
Candidate	320
Confirmed non-lens	4
Uncertain	2
Total	389

$N1$ contains the remaining candidates; $N2$ uses proxies for lens-like negative morphology and is assigned by the first matching condition: DEV type with r-band flux at least 5 nmgy; SER or DEV type with Tractor `shape_r` $\geq$ 1.8 arcsec; $g - r \leq 0.4$ and $r \leq 20.5$ mag; or `shape_r` $\geq$ 2.0 arcsec and r-band flux at least 3 nmgy. These proxies stratify sampling; they are not eligibility cuts and do not use detector scores. Within exposure-count and Tractor-type strata, the procedure targeted an 85:15 $N1 : N2$ sample before final eligibility filtering.

The sampler drew 453,100 rows. Of these, 6,207 lacked a final cutout, leaving 446,893 rows with usable pixels. The final negative pool additionally required finite g, r, z photometry and removed any row within 11 arcsec of a Lenscat system or catalogue-positive centre. A total of 37,342 usable-cutout rows failed at least one of these final photometry or proximity rules. The retained pool therefore contains 409,551 catalogue-negative rows: 343,469 in $N1$ and 66,082 in $N2$. The final stratum proportions need not equal the sampling target because eligibility was applied after sampling. These are catalogue negatives, not individually verified non-lenses.

The positive and negative parent samples were not selected under identical eligibility rules. In particular, the magnitude, morphology, exposure, mask, and declination restrictions above were not applied to the positive catalogue. Eight of the 63 confirmed lenses, and 2,291 of all 4,788 catalogue-positive rows, lie outside the negative sample's declination window. The observed–injected comparison therefore includes differences in deflector populations and survey conditions alongside source insertion.

2.5 Development and test partitions

The clean sample contains 409,551 negatives and 63 confirmed lenses. Before noisy candidate labels are added, the grouped split assigns 286,679 negatives to training, 61,437 to validation, and 61,435 to test; all 63 confirmed lenses are assigned to test before model development. No noisy candidate label is counted as confirmed.

Rows were grouped by object identity, sky proximity, overlapping cutouts, NSIDE= 128 NESTED HEALPix cell (Górski et al. 2005), and known physical system before partitioning. These grouping rules keep objects linked by the recorded identities and spatial or physical-system information within the same partition. Candidate labels were added under the same grouping rules.

Before detector scoring, disjoint spatial groups supplied 20,002 negatives for threshold calibration and 12,512 for false-positive-rate evaluation. The 63 confirmed systems were excluded from detector fitting, validation, checkpoint selection, and threshold calibration. Twenty had an exploratory feature-analysis-development role and are excluded from all final endpoint and observed-reference denominators. The published roster of the remaining 43 systems defines the selected-system endpoint and the observed-reference sample. We do not have a documented general scientific eligibility rule that independently selects those 43 from all 63 confirmed systems. The 43-system roster was not chosen using detector scores. No development object shares a recorded identity or physical group with the calibration, evaluation, or injection samples. Where a spatial-component identifier is available, it also shows no shared component; the calibration, achieved-FPR, and injection-target samples additionally have no flagged development HEALPix or 38-arcsec contact. The available identities, group labels, and sky-neighbour checks show no recorded overlap between development and evaluation samples. Exact footprint polygons are unavailable for part of the development set, so broader geometric overlap cannot be tested. The selected-system recovery fraction is a finite-cohort summary of these 43 heterogeneous systems.

2.6 Positive examples and training-set composition

Because the clean training and validation partitions contain no confirmed positive examples, detector training uses noisy catalogue-positive labels. The 320 objects assigned candidate status after literature review provide one source of positive examples; the 4,399 lower-confidence catalogue candidates provide an additional source. Rows spatially linked to the test partition are excluded before the development sets are formed. This removes 18 reviewed candidates and 290 additional catalogue candidates.

The primary training set combines reviewed and additional catalogue candidates; the comparison set uses reviewed candidates only, with the same negatives (Table 1). Their training negative-to-positive ratios are approximately 79:1 and 1205:1, respectively. Confirmed lenses, confirmed non-lenses, and uncertain systems enter neither composition.

Both positive-example sets contain candidate labels rather than uniform confirmed truth; the contamination fraction of the additional catalogue candidates has not been measured. In natural-prevalence runs, training rows are shuffled and sampled without replacement each epoch, with the incomplete final mini-batch omitted; the loss is unweighted binary cross-entropy. Accuracy is not used for checkpoint selection or operating point assessment because an all-negative classifier would be favoured by the class imbalance while recovering no labelled positives.

2.7 Primary detector protocol

The finder is an ImageNet-initialised EfficientNetV2-S with a binary output (Deng et al. 2009; Russakovsky et al. 2015; Tan & Le 2021), subsequently trained for lens classification. ImageNet-1K is a labelled natural-image dataset whose pretrained weights supply a general visual initialization. The trained finder's pooled representation and the distinct external feature protocols are defined in Section 4.4.

The primary run uses reviewed and additional catalogue candidates, seed 42, natural-prevalence sampling, and a cosine learning-rate schedule. It trains for 160 epochs with AdamW, a five-epoch learning-rate warm-up, and the backbone frozen for the first five epochs. The initial learning rate is 3.88×10^{-4}, and weight decay is 10^{-4}. Training uses mixed-precision forward passes, micro-batches of 64 accumulated to an effective batch size of 512, and omits the incomplete final mini-batch. Early stopping is disabled. Horizontal and vertical flips are applied independently with probability 0.5, followed by a rotation selected uniformly from 0°, 90°, 180°, and 270°. We use no colour or noise augmentation.

The primary detector is the epoch-160 checkpoint; validation, calibration, and evaluation scores do not select it. All primary-detector evaluations use the same preprocessing and operating threshold defined in Section 4.1.

2.8 Detector-sensitivity training panel

The sensitivity panel contains 12 EfficientNetV2-S runs that share the primary architecture and training duration but vary the positive sample, random seed, sampler, or learning-rate schedule:

- six baseline runs cross the two positive-example sets—reviewed candidates only, or reviewed plus additional catalogue candidates—with seeds 42, 123, and 2026 under natural-prevalence sampling and cosine scheduling;
- two sampler runs use seed 42 and seeded 50:50 class-balanced weighted sampling with replacement for the two positive-example sets;
- two schedule runs use seed 42 and natural-prevalence sampling for the two positive-example sets, with a step schedule that halves the learning rate after epoch 130 instead of cosine decay; and
- two additional runs use reviewed candidates only, with seeds 7 and 2027 under the baseline sampler and schedule.

The repeated seeds probe sensitivity to training stochasticity; they are not a statistical sample of all possible trained models. Sampler and schedule pairs change one training choice at seed 42. Changing the positive sample also changes its composition, prevalence, and size; that comparison measures detector sensitivity to the combined change.

For each sensitivity run, we select the checkpoint with the highest validation average precision; selected epochs range from 17 to 103. These checkpoints form a secondary analysis; the epoch-160 checkpoint remains the primary detector. Appendix A6 reports the complete panel.

Figure 1 summarizes the sample, training, and evaluation roles.

3 INJECTION MODELS AND EXPERIMENT SETS

3.1 Overview of simulation and control sets

We use *deflector* for the foreground lens galaxy in an observed strong lens and *injection target* for a catalogue galaxy into whose observed DR10 cutout a simulated lensed source is inserted. The source family, injection-target population, parameter distribution, noise treatment, detector, and operating threshold are distinct design axes that can differ between experiments. Table 3 summarizes each experiment's question, population, image construction, and analysis role. Source, target, noise, and detector settings are specified separately for each experiment. Below, signal-to-noise ratio is abbreviated as S/N and false-positive rate as FPR. We write $p = \mathrm{sigmoid}(\ell)$ for display of a detector's single output logit ℓ; p is not assumed to be a calibrated probability.

The recovery set G_{rec} spans 11 Einstein radii from 0.5 to 3.0 arcsec in 0.25-arcsec steps. Its nine retained r-band PSF–depth cells each contain 500 distinct targets, and every target is rendered at all 11 radii. Source, lens-nuisance, colour, and normalized source-offset draws are shared across radii; only θ_{E} and the corresponding physical source offset vary. The PSF and depth values define descriptive strata, not interventions.

The experiment retains configurations across the full grid without a signal-to-noise or visibility threshold, allowing both recovery and failure to contribute to the conditional response. Population weighting would require an external distribution over the retained cells.

G_{domain} reuses a G_{rec} subset, while G_{match} reuses one target's bare and 1.5-arcsec injected pixels per reference. The primary detector endpoints and G_{rec}, P_{noise}, P_{src}, and C_{bright} use the common primary threshold. The domain and matched analyses instead use continuous cross-fitted readouts in the four feature protocols.

For C_{bright}, the retained 1.5-arcsec payload for target j is decomposed exactly in float32 into bare pixels B_j and non-negative source expectation S_j. If its original intrinsic magnitude is m_j, then at $m \in \{18.5, 19.5, \ldots, 25.5\}$

$$
\begin{aligned}
S_{jm} &= 10^{-0.4(m-m_j)} S_j, \\
X^{\mathrm{E}}_{jm} &= B_j + S_{jm}, \\
X^{\mathrm{M}}_{jmd} &= B_j + \frac{\mathrm{Poisson}(150 S_{jm})}{150}.
\end{aligned}
\tag{3}
$$

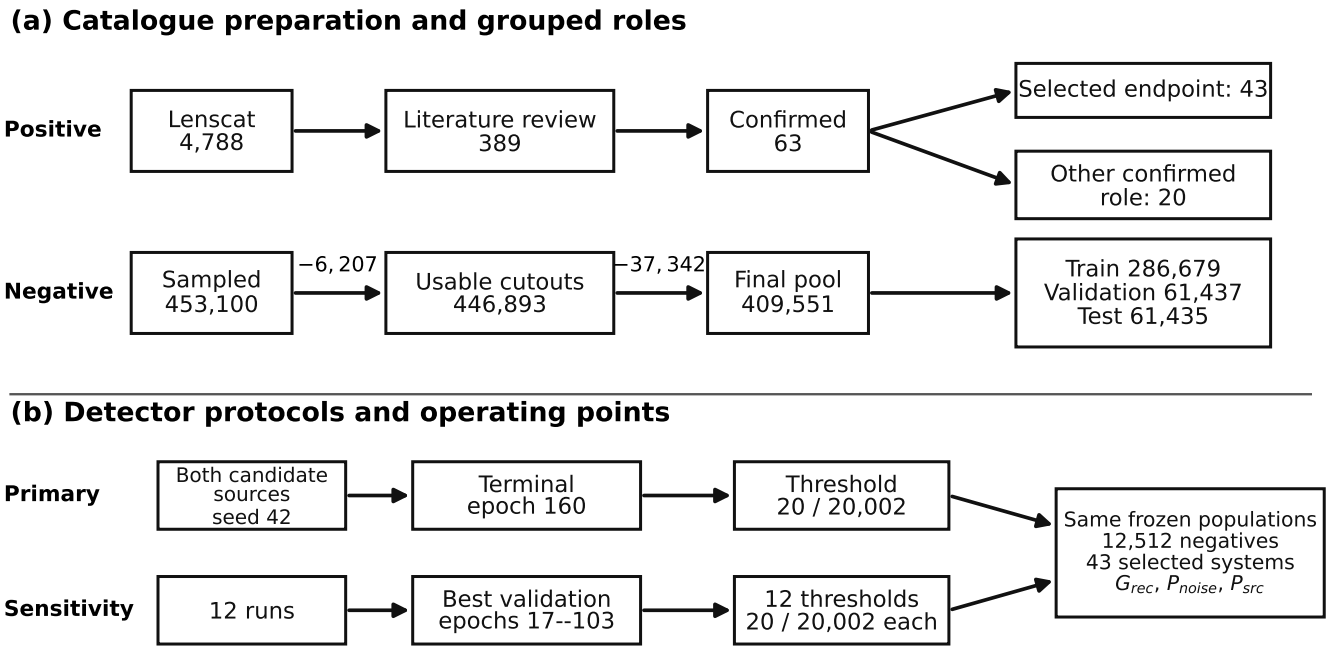


Figure 1. Data preparation, detector training, and evaluation roles. Panel A shows the catalogue-positive evidence review and the negative-sample flow before grouped train, validation, and test assignment. Panel B separates the primary terminal-epoch protocol from the 12-run validation-selected sensitivity panel. The primary detector and each sensitivity checkpoint use the same 20,002-row calibration population. Each model is calibrated independently by choosing the threshold with 20 exceedances, then evaluated on the same corresponding held-out cohorts. The primary uses reviewed and additional catalogue candidates. The sensitivity panel includes both positive-sample compositions and the specified sampler, schedule and seed variants. Arrows show data or model flow; evaluation does not feed back into training.

Alt text: Two linked flow diagrams. The first tracks catalogue positives to 63 confirmed systems and tracks 453,100 sampled negatives to 409,551 retained rows and grouped data partitions. The second shows noisy-label training sets feeding the primary epoch-160 detector and a 12-run sensitivity panel, followed by separate calibration thresholds and the same held-out evaluation samples.

There is one expectation and eight keyed photon realizations per target–magnitude cell. The 3,096 realized images are scored with the detector's training-consistent 20–32-pixel annulus and clip 10. This scales an existing lensed source image and does not repeat lens ray-tracing.

3.2 Injection targets and lens-mass model

Injection targets are catalogue-negative objects from the final negative pool (Section 2.4) with valid 101×101 three-band cutouts and finite cell-assignment quantities. Selection is score-blind and applies the identity, pixel, within-population spatial-group, and physical-group role exclusions in Section 2.5. Distinct observed references and targets can retain cross-population geographic links; these remain in the same probe fold (Section 4.4). Cells use local Tractor `psfsize_r` and `psfdepth_r`, with the latter converted from inverse-square nanomaggy units to 5σ depth. Targets are distinct, but spatial groups can contain multiple targets across cells; the resampling in Section 4.3 preserves these links.

The target roster is selected deterministically without outcomes. Within each retained PSF–depth cell and spatial group, targets are ordered by a namespaced SHA-256 hash of cell and galaxy identifier and assigned a within-group rank. Ranks five and above are removed. The remaining candidates are ordered first by within-group rank and then by hash, and the first 500 form the cell. A cell is retained only when the candidate roster contains at least 500 targets and 100 spatial groups and the five-per-group capped capacity is at least 500.

Within these targets, the operational "massive early-type" catalogue stratum eligible for G_{domain} requires observed z-band magnitude no fainter than 19.5, Sérsic index at least 4, and catalogue axis ratio at least 0.6. These cuts select 576 targets and their 6,336 injected scenes.

For G_{match}, each of the 43 selected confirmed systems is assigned one unique G_{rec} target by the score-blind matching procedure in Section 4.5. In contrast, P_{src} uses the balanced fixed-scene design in Section 3.6, without matching its targets to observed systems.

Every retained rendering uses a singular isothermal ellipsoid (SIE) (Kormann, Schneider & Bartelmann 1994) with external shear (Keeton 2001). The SIE axis ratio is drawn from $\mathcal{U}(0.5, 1.0)$, its position angle from $\mathcal{U}(0, \pi)$, the two shear components independently from $\mathcal{N}(0, 0.05^2)$, and the two offsets of the mass centre from $\mathcal{N}(0, (0.05\ \ \mathrm{arcsec})^2)$. Numerical treatment near the circular SIE limit and its validated support are described in Section 3.7. Einstein radius is the controlled grid axis in G_{rec}, P_{noise}, and P_{src}. It is fixed at 1.5 arcsec in G_{match}.

The SIE parameters are assigned independently of an injection target's light, redshift, velocity dispersion, stellar mass, and environment; axis ratio and orientation are likewise independent of the observed galaxy shape. Table 4 reports actual catalogue coverage before any matching. The selected systems are substantially brighter and larger in the central Tractor fit than the G_{rec} targets. Only 4/4 500

Table 3. Simulation and control sets used in the study. The visibility column states the admission policy; absence of an independent visibility cut retains low-signal configurations. Derived and paired samples reuse the targets, sources, and scene components indicated in the table.

ID	Question	Source and injection targets	Size	Visibility policy	Analysis role
G_{rec}	How often does the primary detector respond over the fixed radius and observing-condition grid?	Single Sérsic sources in catalogue-negative injection targets; source and lens nuisance draws are shared over radius	49 500 injections; 4 500 distinct targets; nine cells	No independent S/N or visibility cut	Fixed-detector recovery and radius curve
G_{domain}	Can the selected observed and injection populations be separated in a fixed representation?	Massive-early-type subset of G_{rec}, compared with the 43 selected confirmed systems	6 336 injections in 576 targets plus 43 observed systems	Inherited from G_{rec}	Dependency-grouped finite-cohort separation
G_{match}	How does a fixed injection move a matched target along an observed–bare feature axis?	Each of the same 43 observed systems and one score-blind matched G_{rec} target, using its bare cutout and corresponding $\theta_{\mathrm{E}} = 1.5$-arcsec injected image	43 triplets (129 rows) in 43 dependency components	No post-render selection or target-flux floor	Cross-fitted representation-axis movement
P_{noise}	How does modelled source-photon sampling change primary-detector response?	Each G_{rec} source expectation and eight $g = 150\,\mathrm{e^-\,nmgy^{-1}}$ Poisson realizations on the same target	49 500 deterministic expectation scenes and 396 000 photon-draw realizations	Inherited from G_{rec}; no independent S/N cut	Photon-minus-expectation paired risk difference
P_{src}	How does the declared structured source construction compare with its single-Sérsic twin?	Restricted high-S/N COSMOS F814W constructions and deterministic catalogue Sérsic twins paired on the same G_{rec} target, with a distinct target for each pair	1 584 pairs; eight draws per pair; 12 672 paired draw comparisons (25 344 arm rows)	Source-quality restrictions only; no post-render S/N or score cut	Photon-averaged paired risk difference
C_{bright}	How does response vary with source brightness and photon realization?	The 43 G_{match} targets with their stored 1.5-arcsec isolated source expectations, rescaled to eight magnitudes	3 096 derived images and training-preprocessing scores	No visibility cut; fixed target and geometry	Fixed-target brightness and source-photon response

targets (0.09 per cent) lie jointly within the selected-system interquartile ranges of z-band magnitude, half-light radius, and Sérsic index. These central-object measurements can include lensed-source light in the selected systems and therefore describe the measured central objects rather than deblended deflectors.

Redshifts are too sparse to inform mass assignment, so the experiment represents a specified image-generation distribution rather than a calibrated deflector population. Complex mass structure, line-of-sight perturbers, and multiple deflectors are not varied as separate factors. Matching uses the five recorded central-object and observing quantities specified in Section 4.5.

3.3 Source models

The parametric source in G_{rec} is a single elliptical Sérsic profile (Sérsic 1968), with the approximation of Ciotti & Bertin (1999), $b_n = 2n - 1/3$, used for the Sérsic constant. Its unlensed r-band magnitude is drawn uniformly from 23 to 26 mag, Sérsic index from $\mathcal{U}(0.5, 2.0)$, effective radius from $\mathcal{U}(0.15, 0.50)$ arcsec, axis ratio from $\mathcal{U}(0.3, 1.0)$, and position angle from $\mathcal{U}(0, 2\pi)$. The source offset is expressed as $\beta_{\mathrm{frac}} = \beta/\theta_{\mathrm{E}}$ and drawn uniformly in source-plane area,

$$U_\beta \sim \mathcal{U}(\beta_{\mathrm{lo}}^2, \beta_{\mathrm{hi}}^2), \qquad \beta_{\mathrm{frac}} = \sqrt{U_\beta}, \tag{4}$$

with $(\beta_{\mathrm{lo}}, \beta_{\mathrm{hi}}) = (0.10, 0.40)$. The same underlying source and lens draw is inherited without alteration by G_{match} and P_{noise}. P_{src} inherits the target, lens mapping, source position, and intrinsic per-band flux but replaces these parametric source-shape parameters with its paired source constructions. Independently of the profile's position angle, the polar angle of the source-centre offset is drawn uniformly from $\mathcal{U}(0, 2\pi)$. The offset is scaled by the nominal θ_{E}, not by the realised caustic of the SIE-plus-shear mapping. Together, these source-position and lens-model priors determine whether a scene forms a double, quad, or ring.

The parametric colours follow $g - r \sim \mathcal{N}(1.15, 0.30^2)$ and $r - z \sim \mathcal{N}(0.85, 0.20^2)$. These values were chosen using observer-frame annular photometry from 388 of the 389 literature-reviewed catalogue records. The measured annular means and standard deviations were 1.23 and 0.39 in $g - r$, and 0.95 and 0.27 in $r - z$; the adopted values are design priors rather than direct population estimates. The catalogue is heterogeneous and the annuli can contain deflector light.

For P_{src}, the structured source-plane construction is drawn from the GalSim COSMOS 25.2 F814W real-galaxy sample (Scoville et al. 2007; Koekemoer et al. 2007; Mandelbaum et al. 2012; Rowe et al. 2015).[3] Eligibility requires elliptical-Gaussian catalogue S/N of at least 100, mask distance of at least 15 native pixels, $17.0 \leq$ `MAG_AUTO` ≤ 24.5, circularised half-light radius 0.12–0.60 arcsec, Sérsic index 0.5–2.0, axis ratio 0.2–1.0, a viable finite single-Sérsic fit, and measured native inscribed support of at least six circularised half-light radii. For the COSMOS 25.2 fit catalogue, the single-Sérsic status is the fifth component (zero-based index 4) of `FIT_STATUS`; codes 0 and 5 are excluded. No detector output enters eligibility or selection.

The structured profile uses the GalSim `RealGalaxy` deconvolution of the observed F814W stamp, with no noise padding, fol-

[3] The archived dataset is Mandelbaum et al. (2012), `https://doi.org/10.5281/zenodo.3242143`.

Table 4. Pre-match catalogue coverage of the final $G_{\rm rec}$ injection targets and the 43 selected confirmed systems. Numerical rows give the median [interquartile range] over finite values. The listed quantities are complete for all 4,500 injection targets. Tractor type counts are SER/DEV/REX/other; the one selected-system "other" object is EXP. These are central-object catalogue measurements and can include lensed-source light.

Catalogue quantity	$G_{\rm rec}$ targets (N = 4 500)	Selected confirmed systems (N = 43)
Tractor type counts	2 853/1 213/434/0	33/9/0/1
r magnitude	20.17 [19.44, 20.84]	18.16 [17.49, 19.79]
z magnitude	19.40 [18.71, 19.77]	17.11 [16.59, 18.32]
Half-light radius (arcsec)	1.05 [0.73, 1.45]	5.48 [3.63, 7.37]
Sérsic index	2.60 [1.00, 4.00]	4.34 [3.68, 5.68]
Axis ratio	0.61 [0.43, 0.78]	0.70 [0.62, 0.80]

lowed by reconvolution to a common circular Gaussian of 0.15-arcsec FWHM. This scale is broader than the original F814W PSF for every eligible source. The deterministic twin uses the same object's catalogue Sérsic index, circularised half-light radius, axis ratio, and position angle and is convolved to the same 0.15-arcsec resolution. Both receive the same source-specific random rotation. They are drawn on a 161×161 grid with 0.03-arcsec pixels.

After the common low-pass operation, both maps are clipped to non-negative surface brightness, shifted to the geometric grid centre with a 3×10^{-4}-arcsec tolerance, and zeroed outside the same circular support. That support is 95 per cent of the measured native inscribed radius, capped at 2.35 arcsec. Each arm must retain at least 90 per cent of its pre-projection positive flux; the allowed pre-projection negative-to-positive flux ratios are 0.20 for the structured map and 0.001 for the Sérsic map. Each projected map is separately normalised to unit discrete source-plane integral.

The 770 eligible sources are stratified by a 4×4 crossing of circularised half-light-radius and single-Sérsic-fit-residual quantiles. The residual variable is the catalogue-provided `FIT_MAD_S` statistic for the single-Sérsic fit. Within each stratum, a namespaced SHA-256 ordering is computed from the COSMOS identifier. Ranks 0–8 form the 144-source experiment set and ranks 9–17 form the disjoint 144-source design set. One F814W spatial map is constructed for each arm and scaled by the inherited intrinsic flux in each of g, r, z. Thus each arm reuses its own spatial map across the three bands. The construction is therefore achromatic, and the structured arm retains the F814W measurement uncertainty propagated through its preparation.

3.4 Ray tracing, flux, PSF, and image construction

For each injection, we evaluate

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}_{\rm SIE}(\boldsymbol{\theta}) - \boldsymbol{\alpha}_{\rm shear}(\boldsymbol{\theta}) \tag{5}$$

where the SIE deflection includes the stated mass-centre offset and the shear deflection uses the two sampled shear components. Gravitational lensing conserves surface brightness, not total flux (Narayan & Bartelmann 1996). The source model is normalised to its specified unlensed flux, its surface brightness is evaluated at the ray-traced source-plane positions, and the image-plane flux is allowed to change under the resulting magnification; the lensed image is never renormalised.

For $G_{\rm rec}$, tensor Gauss–Legendre quadrature integrates the analytic Sérsic surface brightness over each native 0.262-arcsec pixel. The analytic elliptical-Sérsic integral sets the unlensed per-band flux. Orders 24 and 32 provide the registered per-scene comparison: the unlensed-flux relative error must be at most 0.01, the successive lensed-image relative L_1 difference at most 0.005, and the relative peak difference at most 0.02. One retained scene required the score-blind order-48 extension.

We approximate the local PSF in each band by a circular Gaussian. Its r-band FWHM is taken from the injection target's Tractor `psfsize_r` value; the g- and z-band FWHM values are set to 1.05 and 0.94 times the r-band value, respectively. These fixed ratios are renderer approximations rather than per-target measurements of the g- and z-band PSFs. The rendered lensed source is converted to nanomaggies per native pixel and convolved after pixel integration on the native grid. The Gaussian kernel is truncated at 8σ, and explicit zero padding prevents periodic wrap-around. The renderer therefore uses a circular effective post-pixel coadd PSF; exposure-level optical structure and PSF anisotropy are outside this model. The result is added to the observed injection-target cutout before the preprocessing in Section 2.2. The target cutout therefore remains the same within a controlled pair.

The $G_{\rm match}$ injected member is the exact stored $\theta_{\rm E} = 1.5$-arcsec $G_{\rm rec}$ deterministic expectation array, and its bare member is the exact stored target cutout; neither arm is rerendered or flux-rescaled. For $P_{\rm src}$, each non-parametric source map is bilinearly interpolated with zero outside its recorded support on a grid oversampled by a factor of eight in each image dimension. Surface brightness is block-averaged to native pixels, multiplied by the fixed inherited intrinsic per-band flux and native-pixel area, and then convolved with the same zero-padded effective post-pixel Gaussian. The 4×, 8×, and 16× renderings of every final pair must have consecutive relative L_1 differences below 0.01; the maximum in the final bank is 0.0022. These comparisons assess numerical integration at the fixed DR10 pixel scale and effective PSF.

3.5 Noise model

An observed DR10 injection-target cutout already contains foreground light and realised sky, detector, coaddition, and survey-processing noise. The stored $G_{\rm rec}$ expectation adds rendered source light without an additional source-photon or sky-noise field. $P_{\rm noise}$ and $P_{\rm src}$ add only the modelled source-photon realization; neither redraws the target's existing noise.

For non-negative injected-source expectation I in nanomaggies per native pixel and a modelled conversion g in electrons per nanomaggy, $P_{\rm noise}$ draws

$$E = g \max(I, 0), \qquad E' \sim \text{Poisson}(E),$$
$$I_{\rm photon} = \frac{E'}{g} = I + \frac{E' - E}{g}. \tag{6}$$

Pixels with zero injected flux therefore receive zero source counts. We use $g = 150\ \mathrm{e^-\ nmgy^{-1}}$ as a modelled conversion, not as a calibrated gain for every contributing DR10 exposure. One independent PCG64DXSM stream is derived by SHA-256 from each scene and draw identity. Every deterministic expectation scene receives eight draws, each added to the exact stored target bytes.

A separate pilot, conducted without detector scores, was used to choose the number of photon realizations. Appendix Table A5 reports the pilot sample, the tested draw counts, and their Monte Carlo standard errors (MCSEs).

The P_{src} arms use the same modelled $g = 150$ conversion but couple their Poisson draws. For structured and Sérsic expected electron intensities $\lambda_{\mathrm{str}} = gI_{\mathrm{str}}$ and $\lambda_{\mathrm{ser}} = gI_{\mathrm{ser}}$, let $\lambda_{\mathrm{min}} = \min(\lambda_{\mathrm{str}}, \lambda_{\mathrm{ser}})$ pixelwise. The construction draws

$$\begin{aligned} &C \sim \mathrm{Poisson}(\lambda_{\mathrm{min}}), \\ &A \sim \mathrm{Poisson}(\lambda_{\mathrm{str}} - \lambda_{\mathrm{min}}), \qquad B \sim \mathrm{Poisson}(\lambda_{\mathrm{ser}} - \lambda_{\mathrm{min}}), \\ &I'_{\mathrm{str}} = (C + A)/g, \qquad I'_{\mathrm{ser}} = (C + B)/g, \end{aligned} \tag{7}$$

using three independent keyed streams. Each arm retains its exact Poisson marginal; equal expected intensities produce identical counts, while only the excess intensity is drawn independently.

The F814W measurement uncertainty retained by the structured P_{src} map is distinct from the source-photon draw.

Figure 2 illustrates both source constructions at the expectation and photon-realized stages.

3.6 Paired source construction

The final P_{src} design has 99 target-cell–radius blocks. Each block contains 16 distinct targets, with four from each within-cell intrinsic-r-magnitude quantile and one assignment from each source stratum. This gives 1,584 pairs on distinct targets; each of the 144 empirical sources is used 11 times.

Targets are ordered by a namespaced SHA-256 hash within each cell and intrinsic-magnitude quantile before four are assigned to each radius. Parent scenes are then ordered within cell–radius blocks, and the 16 source strata are cycled deterministically across those blocks using the hash-ordered source rosters. The assignment uses no detector score, threshold decision, or probe feature.

Within a pair, the structured and deterministic single-Sérsic maps share the exact target, lens and shear parameters, source position, aligned source-plane centroid, effective per-band PSF, intrinsic $g/r/z$ fluxes, renderer, and photon draw indices. The primary design fixes intrinsic flux before lensing and applies no post-lensing rescaling. Differences in magnification induced by the two source maps therefore remain part of the source-construction intervention. The realised flux centroids are not explicitly aligned after rendering.

A separate design pilot uses disjoint targets and sources under the same cell, radius, magnitude, and source-stratum structure. Eight is the smallest tested draw count meeting its paired-risk MCSE criterion; Appendix Table A13 gives the candidate panel.

The comparison in Section 4.8 therefore isolates the two complete source-construction pathways at the same intrinsic flux and nominal source-plane centre.

3.7 Renderer validation

A stratified 142-scene audit evaluates quadrature, an independently refined integration, an effective-PSF reference, and SIE deflections from Lenstronomy (Birrer & Amara 2018). Appendix A15 reports its support and numerical bounds. These tests assess numerical accuracy for the configurations examined; they do not test whether the adopted source, lens, PSF, or detector models are astrophysically complete.

4 ANALYSIS DESIGN AND STATISTICAL METHODS

This section defines the operating points, estimators, and uncertainty procedures used for the experiment sets in Table 3.

4.1 Evaluation cohorts, scoring procedure, and operating points

The primary selected-system endpoint uses the 43-system evaluation cohort defined in Section 2.5. None was used for detector fitting or validation, detector-model selection, threshold calibration, or exploratory feature-analysis development. The 20 systems assigned to that exploratory role are excluded from this denominator.

Two disjoint catalogue-negative samples are drawn as whole spatial groups from the detector's negative test population. The threshold sample contains 20,002 rows in 1,768 spatial groups. A second sample contains 12,512 rows in 1,089 spatial groups and is used only to measure the achieved FPR without modifying the calibrated threshold.

For the primary analyses, the selected-system endpoint, G_{rec}, P_{noise}, and P_{src} use the same epoch-160 EfficientNetV2-S checkpoint, preprocessing, and scoring procedure. The network is in evaluation mode: all batch-normalisation layers use stored running statistics and dropout is inactive. Inference uses float32 operations. Logits define the operating point because extreme float32 sigmoid values can saturate numerically. The displayed $p = \mathrm{sigmoid}(\ell)$ is not treated as a calibrated probability. Execution-environment and batching details are reported in Appendix A.

For N_0 calibration negatives with native float32 logits sorted in descending order $s_{(1)} \geq \cdots \geq s_{(N_0)}$, a target FPR α permits $k = \lfloor \alpha N_0 \rfloor$ exceedances. We choose

$$\begin{aligned} t_\alpha &= \mathrm{nextafter}_{32}\left(s_{(k+1)}, +\infty\right), \\ D_i(t_\alpha) &= \mathbb{I}(\ell_i \geq t_\alpha). \end{aligned} \tag{8}$$

With $N_0 = 20\,002$, $\alpha = 10^{-3}$, and $k = 20$, this gives $t_{10^{-3}} \simeq 1.572$ and 20 calibration exceedances. The full-precision float32 value is reported in Appendix A. The threshold is applied unchanged to every primary detector-response analysis; the feature probes do not use it.

4.2 Detector-checkpoint sensitivity

The primary detector is the seed-42 epoch-160 checkpoint trained with reviewed and additional catalogue candidates. The sensitivity analysis selects the highest-validation-AP checkpoint from each of the 12 training runs, retaining the earlier epoch under ties. Each checkpoint receives its own threshold from the same calibration sample using Equation 8, and that threshold is applied to the common evaluation data. The target 10^{-3} defines the empirical calibration rate. Checkpoint-specific estimates and intervals condition on t_m.

The pixels, estimators, and dependence-aware resampling procedures are unchanged across checkpoints; only model weights, scores, and thresholds vary. The operational G_{domain} and G_{match} probes characterize only the designated primary checkpoint and are not repeated across the panel. The panel tests how thresholded detector responses change across trained models; the feature-space analysis is performed only for the primary detector. The 12 checkpoints are correlated variants within one detector design (Schaefer et al. 2018; Cañameras et al. 2024). We report both validation-selected and terminal-epoch checkpoints; comparisons between these checkpoint policies are exploratory.

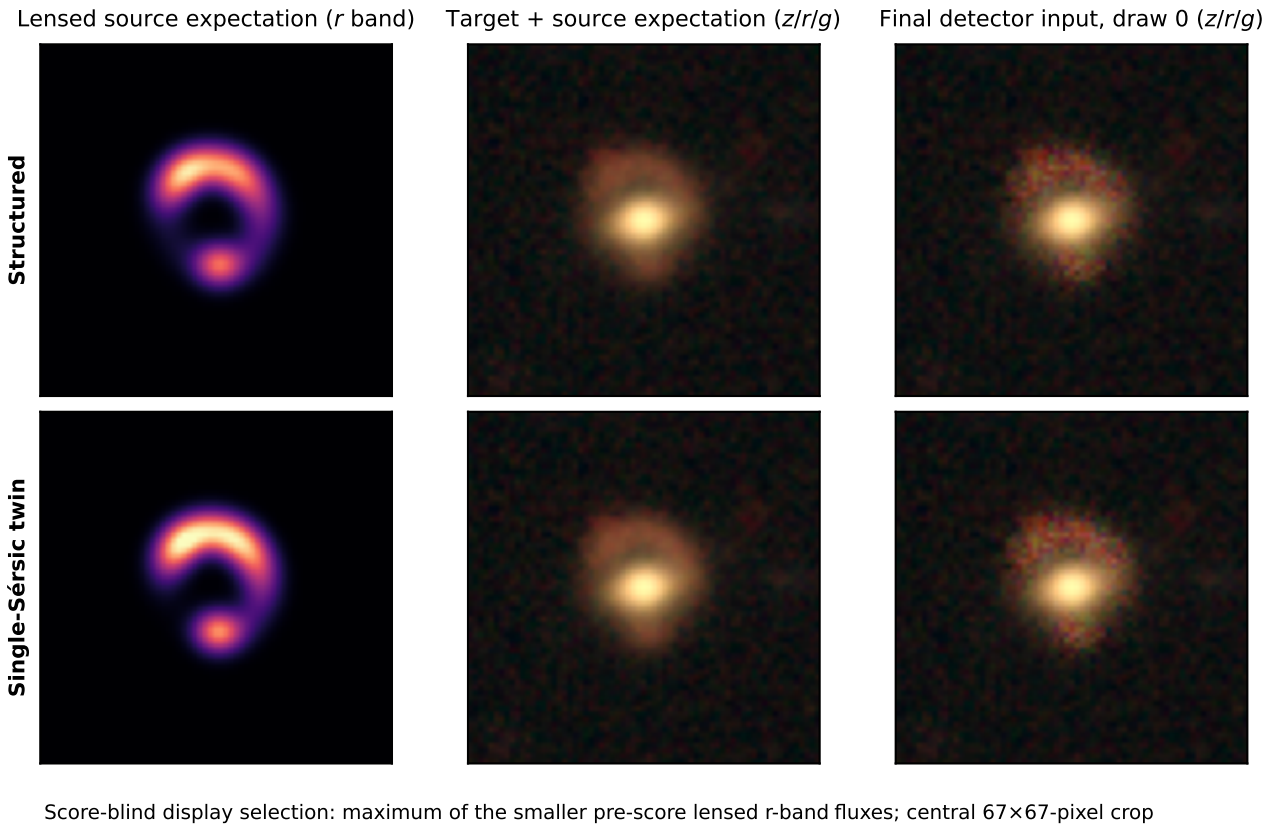


Figure 2. Illustration from the final $P_{\rm src}$ bank. The pair is selected without detector scores by maximizing the smaller of the two pre-score lensed r-band fluxes, solely to make the inserted light visible. This display selection does not alter the analysis population; the pair remains in the final paired analysis. Rows show the structured construction and its deterministic single-Sérsic twin. Columns show the lensed source expectation in r, the target plus source expectation in $z/r/g$ colour, and final photon draw 0 supplied to the detector. The pair shares the target, source identity, lens mapping, nominal source-plane centre, effective PSF, intrinsic per-band flux, and photon coupling, but receives no post-lensing flux rescaling. Panels use the central 67×67-pixel display crop; the detector uses all 101×101 pixels.

Alt text: Two rows compare the structured source and its smooth single-Sérsic twin. The first column shows similar but visibly non-identical ring-like r-band source expectations. The second shows each expectation added to the same bright central survey galaxy. The third shows photon draw 0, where coupled source-photon fluctuations are visible around the shared foreground scene.

4.3 Detector-performance and recovery estimators

For a labelled population at a stated threshold, recall is TP/(TP+FN), FPR is FP/(FP + TN), and precision is TP/(TP + FP). ROC AUC measures ranking across all thresholds, with half credit for ties. The reported precision–recall summary is average precision. The Brier score is

$$B = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2. \tag{9}$$

Validation-positive metrics use noisy reviewed and additional catalogue candidate labels and are development diagnostics, not estimates on confirmed systems.

For a named injection population Q, recovery at threshold t is

$$\widehat{C}_Q(t) = \frac{1}{|Q|}\sum_{i\in Q}\mathbb{I}(\ell_i \geq t). \tag{10}$$

This quantity is conditional on the target population, source and mass-model priors, observing-condition grid, visibility policy, detector, and threshold specified for Q. We call this quantity conditional recovery rather than survey-population completeness. Because $G_{\rm rec}$ imposes no independent detectability or visibility condition, its recovery fraction combines configurations that may not be physically detectable with the finder's response; it is not an estimate of finder response conditional on detectability alone.

For x row-level successes among n rows, the 95 per cent Wilson score interval (Wilson 1927) is

$$p_{\rm W} = \frac{\widehat{p} + z^2/(2n)}{1 + z^2/n},$$
$$\Delta_{\rm W} = \frac{z}{1 + z^2/n}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n} + \frac{z^2}{4n^2}}, \tag{11}$$
$$\mathrm{CI}_{\rm W} = [p_{\rm W} - \Delta_{\rm W}, p_{\rm W} + \Delta_{\rm W}],$$
$$\widehat{p} = x/n, \qquad z = \Phi^{-1}(0.975) \simeq 1.96.$$

We apply this row-level interval to the achieved-FPR endpoint. Because nearby catalogue rows can be dependent, that endpoint also receives a 10,000-draw spatial-group interval: each draw assigns one independent mean-one exponential multiplier to each spatial group

and takes the weighted ratio of exceedances to rows. The Wilson and multiplier intervals correspond to different dependence summaries. The 43 selected confirmed systems are reported as an exact finite cohort without a binomial or population interval.

For $G_{\rm rec}$, let D_{chr} be the threshold indicator for target h in retained observing cell c at radius r. With nine cells, 500 targets in each cell, and 11 radii per target, the primary point estimator is

$$\widehat{C}_{\rm rec} = \frac{1}{9}\sum_{c=1}^{9}\frac{1}{500}\sum_{h=1}^{500}\frac{1}{11}\sum_{r=1}^{11} D_{chr}. \tag{12}$$

Thus cells, targets within cells, and the 11 radius nodes receive equal weight. This equals the raw row mean because the design is balanced; it is not a survey-population weighting.

Sampling uncertainty uses 10,000 exponential-multiplier draws at the injection-target spatial-group level. A group receives one global multiplier per draw, and that same multiplier follows all of its targets wherever they occur across cells and follows every radius of each target. Within each cell, the weighted target mean is formed as a ratio. To remove finite-draw ratio bias without changing the joint covariance, the draw distribution for every cell–radius entry is recentered to its exact equal-target point estimate. The overall 95 per cent interval is the 2.5th and 97.5th percentiles of the recentered, equal-cell and equal-radius draws.

The radius curve averages cells and targets at each radius. Its unstudentised simultaneous band uses

$$\begin{aligned} q_{0.95} &= \mathrm{quantile}_{0.95}\left(\left\{\max_r\left|\widehat{C}_r^{*(b)} - \widehat{C}_r\right|\right\}_{b=1}^{10000}\right), \\ \mathcal{B}_r &= [\widehat{C}_r - q_{0.95},\ \widehat{C}_r + q_{0.95}]. \end{aligned} \tag{13}$$

The band is not clipped to $[0, 1]$. Every cell–radius estimate has 500 distinct targets; the resampling unit, rather than the injection count, defines independent support.

4.4 Linear probes and grouped evaluation

A linear probe consists of fold-local standardisation followed by L_2-regularised logistic regression on features computed with encoder weights held fixed. Every probe uses $C = 1$, the `lbfgs` solver, `max_iter=5000`, estimator seed 42, and scikit-learn 1.9.1. The weighted mean and variance used by `StandardScaler` are fitted only on the training folds, and the same training weights are supplied to the logistic-regression fit.

Dependency groups are defined without examining feature values. They are the transitive union of shared physical group, NSIDE= 128 NESTED HEALPix cell (Górski et al. 2005), within-population spatial group, reference identity, injection-target identity and underlying-scene identity, source, lens, PSF, renderer and render-parameter identities, random-draw identity, and base or final pixel hash. Every resulting dependency component remains wholly in one of five folds; score-blind fold allocation uses seed 42. This closes indirect links, including a reference geographically linked to an injection target in a different nominal pair.

Across the full set of rows eligible to fit an axis, the two fitting classes receive equal total base weight. Within a class, every dependency component receives equal total weight, divided equally over its rows. Each fold fit uses the corresponding weights of its retained training components without fold-specific renormalisation. For fitting-roster size N, class y represented by K_y components, and $n_{c,y}$ class-y rows in component c, the weight is

$$w_i = \frac{N}{2K_y n_{c,y}}. \tag{14}$$

For $G_{\rm domain}$, the weights sum to 6,379, with 3,189.5 assigned to each class. Consequently, repeated radii from one connected target do not receive eleven times the training weight of a singleton component. The same full-roster component-equal, class-balanced weights define the primary pooled out-of-fold (OOF) ROC AUC. A row-weighted OOF AUC is retained only as a diagnostic. ROC AUC uses half credit for tied positive–negative scores.

The domain probe compares all 43 selected confirmed systems with the exact 6,336-row $G_{\rm domain}$ subset in 576 injection targets. The common dependency graph contains 561 components: 43 contain observed rows, 520 contain simulated rows, and two contain both. There are zero component crossings over the five folds. The comparison uses four feature sets. The operational representation $z(x)$ is the 1,280-component output after the primary detector's global-average-pooling layer and before its dropout and linear classifier. The existing lens head evaluates

$$h(x) = w_{\rm lens}^{\mathsf{T}} z(x) + b. \tag{15}$$

The operational domain probe instead fits fold-specific regularized linear readouts of fold-standardized $z(x)$ to observed-versus-injected labels, using training rows only. This examines the representation that feeds the finder without restricting the diagnostic to its learned lens-classification direction. The existing lens head is reported as a fixed-score baseline alongside the four fitted feature protocols. Both use the detector-native full 101×101 g, r, z input with the trained mapping defined in Section 2.2.

The Stein representation is the 2,048-component global-average-pooled output of its self-supervised ResNet-50 before the projection head (Stein et al. 2022). Its encoder was trained with a contrastive objective on unlabelled Legacy Survey images. Object-level overlap between those pretraining images and the present cutouts is unknown, so the Stein result is interpreted as feature-protocol sensitivity. The generic ImageNet representation is the 1,280-component global-average-pooled output of a separate EfficientNetV2-S, with different weights from the lens-label-conditioned operational detector. The fourth representation contains 18 fixed pixel statistics. Each representation includes its own preprocessing. The two external deep encoders use the Legacy Survey display mapping adopted by Stein et al. (2022). For nanomaggy channels x_c, let $u_c = a_c x_c + m$, with $(a_g, a_r, a_z) = (6.0, 3.4, 2.2)$, $m = 0.03$, and $q = 20$. The exact mapping is

$$\begin{aligned} I &= \frac{1}{3}\sum_{c\in\{g,r,z\}} \max(0, u_c), \\ F(I) &= \frac{\operatorname{asinh}(qI)}{\sqrt{q}\, I_{\rm safe}}, \\ (R, G, B) &= \mathrm{clip}_{[0,1]}\left[(u_z, u_r, u_g)F(I)\right], \end{aligned} \tag{16}$$

where $I_{\rm safe} = I + 10^{-6}\mathbf{1}[I = 0]$. The Stein input is the central 96×96 pixels. The ImageNet input uses all 101×101 pixels, is bilinearly resized with antialiasing to 384×384, and receives the standard ImageNet channel normalisation. The pixel representation records, separately in each g, r, z band, the mean, standard deviation, median, 99th percentile, minimum, and maximum. Probe comparisons therefore diagnose the frozen representation together with its preprocessing, not encoder weights alone.

$G_{\rm domain}$ reports dependency-component-equal pooled OOF AUCs as finite-cohort, model-conditional point estimates. No resampling interval or significance test is attached. We compare all four representations on the same rows, folds, component weights, and observed–injected labels. The five fold-level AUCs show how performance

varies across held-out subsets containing only eight or nine observed systems.

4.5 Unpaired separation and matched controls

We distinguish three questions. An unpaired real–injection probe asks whether the selected populations are separable. A real–bare-target probe measures separation present before an arc is added. A bare–injected comparison holds the target fixed and measures how the added arc moves that target in feature space. We interpret these estimands separately because they answer different population and paired-movement questions.

The matched-baseline set G_{match} uses the exact 43 selected confirmed systems as observed references. They are matched without replacement to the 4,500 G_{rec} injection targets on observed z-band magnitude, catalogue Sérsic index, axis ratio, r-band PSF FWHM, and r-band depth. No detector score or representation feature enters matching.

For each covariate k, the complete 4,500-target sample supplies the median m_k and interquartile range s_k, using all finite values of that component. We define $z_{ik} = (x_{ik} - m_k)/s_k$. All five values must be finite for a target to be eligible; all 4,500 targets meet this requirement, and no imputation is used. A reference–target edge is allowed only when

$$\begin{aligned} &|z_{\mathrm{ref},k} - z_{\mathrm{target},k}| \le 2 \quad \text{for every } k, \\ &\left[\sum_k (z_{\mathrm{ref},k} - z_{\mathrm{target},k})^2\right]^{1/2} \le 3. \end{aligned} \tag{17}$$

Among allowed edges, maximum-cardinality then minimum-total-distance assignment is used, with SHA-256-sorted identifiers breaking ties. All 43 references are matched to unique targets.

The supplementary material includes the complete full-field reference–target atlas, the matching and brightness-parent tables, covariate balance summaries, and the code used to generate the displays.

Each target contributes its exact stored bare cutout and its exact $\theta_{\mathrm{E}} = 1.5$-arcsec G_{rec} deterministic expectation image. There is no rerendering, post-render selection, flux rescaling, imputation, replacement, or target-brightness floor. The resulting 43 observed–bare–injected triplets form 43 closed dependency components.

In each representation and fold, the logistic axis is fitted only to the training-fold observed and bare rows, with observed as the positive direction; injected rows fit neither the axis nor its scale. Let a^x_{jf} denote a held-out decision-function score for target j, fold f, and arm $x \in \{\mathrm{bare}, \mathrm{inj}\}$. We standardise both arms using the weighted mean μ^{bare}_f and population standard deviation σ^{bare}_f of the training-fold bare scores:

$$\begin{aligned} \tilde{a}^x_{jf} &= \frac{a^x_{jf} - \mu^{\mathrm{bare}}_f}{\sigma^{\mathrm{bare}}_f}, \\ M_j &= \tilde{a}^{\mathrm{inj}}_{jf} - \tilde{a}^{\mathrm{bare}}_{jf}, \qquad \widehat{\Delta}_{\mathrm{match}} = \frac{1}{43}\sum_{j=1}^{43} M_j. \end{aligned} \tag{18}$$

Here M_j is an equal-target movement measured in training-bare standard-deviation units toward the observed-reference side of a cross-fitted axis.

Pooled observed–bare, observed–injected, and injected–bare AUCs, with half credit for ties, summarize the same fitted axis. These point summaries condition on the frozen score-blind assignment; no formal interval or significance test is attached. The matching set contains five measured central-object and observing quantities; complete photometric-redshift and source-magnitude covariates are unavailable, and the observed central-object quantities can include lensed-source light.

4.6 Fixed-target brightness and source-photon response

C_{bright} uses equation (3) on 43 fixed targets under the detector's trained 20–32-pixel annulus and clip-10 preprocessing. For each target and magnitude, photon indicators are first averaged over the eight keyed draws; the resulting target-level values are then averaged equally over targets. Primary summaries are recovery and native-logit response over magnitude; photon-minus-expectation is the paired secondary contrast. If $\widehat{R}_d$ is the equal-target photon recovery at one magnitude for draw index d, the finite-draw diagnostic is

$$\mathrm{MCSE}_{\mathrm{bright}} = \frac{\mathrm{sd}(\widehat{R}_1, \ldots, \widehat{R}_8)}{\sqrt{8}}. \tag{19}$$

No population interval or visibility admission rule is attached. Source-only in-stamp flux and peak intensity are diagnostics rather than calibrated arc S/N.

4.7 Arc-level Poisson experiment

P_{noise} compares all 49,500 stored G_{rec} expectations with eight modelled source-photon realizations per scene at $g = 150\,\mathrm{e^-\ nmgy^{-1}}$, using the common threshold. The target and expected arc remain fixed; only source-photon realization changes (Section 3.5).

Let D^{E}_{chr} denote the deterministic expectation decision and D^{M}_{chrd} the modelled-photon decision for draw d. The primary modelled-photon-minus-expectation estimand is

$$\widehat{\Delta}_{\mathrm{noise}} = \frac{1}{9}\sum_{c=1}^{9}\frac{1}{500}\sum_{h=1}^{500}\frac{1}{11}\sum_{r=1}^{11}\left\{\frac{1}{8}\sum_{d=1}^{8} D^{\mathrm{M}}_{chrd} - D^{\mathrm{E}}_{chr}\right\}. \tag{20}$$

Photon draws are therefore averaged within deterministic expectation scene before radii are averaged within target; targets and retained cells then receive equal weight. Repeated photon draws never increase a target's scientific weight.

The sampling interval uses 10,000 target-spatial-group exponential-multiplier draws with the same global grouping, equal-cell weighting, and cell–radius recentering as G_{rec}. Its uncertainty is conditional on the realized target and source construction and does not include gain-model or threshold-calibration uncertainty. The radius-specific effects receive the same maximum-absolute-deviation simultaneous-band construction as equation (13). Gained and lost threshold crossings are reported as descriptive probabilities over the complete scene–draw rows.

For scene i, let s_i^2 be the sample variance of its eight modelled threshold indicators and let $w_i = 1/49\,500$, which is the final equal-cell, equal-target, equal-radius weight under the balanced design. The finite-draw contribution is

$$\mathrm{MCSE}_{\mathrm{noise}} = \left(\sum_{i=1}^{49\,500} w_i^2 \frac{s_i^2}{8}\right)^{1/2}. \tag{21}$$

The estimate applies to the specified stochastic source-photon model, whose $g = 150$ conversion and post-coadd construction define the intervention.

4.8 Paired source estimators

For pair j and photon draw d, let D^{ser}_{jd} and D^{str}_{jd} indicate whether the corresponding photon-realized images from the single-Sérsic and structured constructions exceed the common logit threshold. The underlying source constructions are deterministic, but the scored rows are draw-specific images. The primary estimand is the Sérsic-minus-structured photon-averaged risk difference,

$$\widehat{\Delta}_{\mathrm{src}} = \frac{1}{1584}\sum_{j=1}^{1584}\frac{1}{8}\sum_{d=1}^{8}\left(D^{\mathrm{ser}}_{jd} - D^{\mathrm{str}}_{jd}\right). \tag{22}$$

Operationally, draw differences are first averaged within pair; pairs are then averaged within each of the nine fixed target cells and cells are averaged equally. The expression above is identical because every cell has the same number of pairs, every cell–radius block contains all 16 source strata, every selected source is used 11 times, and all base analysis weights are one.

The 95 per cent interval is a 10,000-draw multiway exponential-multiplier interval. Each draw assigns one global mean-one exponential multiplier to each of the 1,219 target spatial groups and applies it to that group's rows in every cell and radius. It independently assigns one multiplier to each of the 144 empirical sources; source multipliers are normalised to mean one within each of the 16 fixed source strata. The product of the spatial and source multipliers weights each pair, after which equal-cell ratio estimates are averaged. Draws are recentered to $\widehat{\Delta}_{\mathrm{src}}$, and the 2.5th and 97.5th percentiles form the reported interval. This procedure preserves dependence from repeated sources and geographically linked targets. The resulting interval is a design-based approximation, not an exact binary-outcome interval.

Finite-photon-draw Monte Carlo uncertainty is reported separately. Let $\widehat{\Delta}_d$ be the same equal-cell risk difference computed from draw index d alone. Then

$$\mathrm{MCSE}_{\mathrm{src}} = \frac{\mathrm{sd}(\widehat{\Delta}_1, \ldots, \widehat{\Delta}_8)}{\sqrt{8}}. \tag{23}$$

The point estimate and multiplier interval condition on the calibrated threshold. The empirical construction retains measurement uncertainty, making the estimand a comparison of the complete source-construction pathways.

4.9 Interpretation of estimates and intervals

The achieved-FPR, G_{rec}, P_{noise}, and P_{src} intervals use the 95 per cent procedures defined above. The selected 43-system endpoint and both feature-probe analyses are finite-cohort point summaries without population intervals. The four feature protocols form one finite-cohort comparison. The operational probes characterize the primary epoch-160 checkpoint; the 12-checkpoint panel evaluates thresholded detector-response sensitivity and does not repeat the feature probes.

Each thresholded estimate conditions on the threshold calibrated for its checkpoint. Multiplier intervals describe the specified target/source dependence but do not incorporate detector-training, threshold-calibration, gain-model, source reconstruction, or population-transport uncertainty. Finite-photon MCSE is reported separately from sampling uncertainty. Formal equivalence assessment would require a predefined margin beyond these estimation summaries.

5 RESULTS

5.1 Validation context and disjoint endpoints

The operating threshold was calibrated on 20,002 catalogue-negative examples before endpoint scoring. At $t_{10^{-3}} = 1.572$, 20 calibration examples met or exceeded threshold; Appendix A gives the full-precision float32 value and comparison rule.

In the separate achieved-FPR endpoint, 13 of 12,512 catalogue-negative rows exceed $t_{10^{-3}}$, or 0.104 per cent. The row-Wilson 95 per cent interval is [0.061, 0.178] per cent, and the spatial-group multiplier interval is [0.055, 0.166] per cent. This is the threshold-exceedance rate for the fixed evaluation cohort at the previously calibrated operating point.

The detector also responds to 39 of the 43 selected confirmed systems (90.7 per cent). We report this selected-cohort fraction without a population interval because the systems combine heterogeneous discovery programmes, catalogue eligibility, and galaxy- and cluster-scale lenses. Section 5.7 reports this endpoint for the 12 sensitivity checkpoints. Appendix A reports the separate noisy-label development diagnostics; threshold calibration and selected-system evaluation use the cohorts defined above.

5.2 G_{domain} linear-probe separation

Observed references and simulated injections are distinguishable in all four feature protocols (Table 5). Cross-fitted domain-ranking AUC is 0.991 in the operational detector's pooled penultimate representation, 0.958 in Stein self-supervised survey features, 0.979 in generic ImageNet features, and 0.981 in pixel statistics. Thus strong separation appears in the lens-trained representation, external learned representations, and elementary image summaries.

The existing lens-classification head also separates these populations strongly (Table 5), showing that the difference is expressed in the operational score as well as the fitted penultimate readouts. The two readouts share a feature vector; no significance test compares their AUCs. Each finite-cohort AUC describes its feature protocol, including its training history, preprocessing, and fitted readout. Native-score summaries appear in Appendix A2.

5.3 G_{match} shared-axis diagnostics

The score-blind G_{match} design contains 43 triplets: each selected observed reference, one matched bare G_{rec} target, and that target with its fixed 1.5-arcsec injection. The cross-fitted axis in each representation is learned from observed-reference and bare-target rows only; injected rows are not used to fit the axis or its scale. There are 129 rows in 43 dependency components. Appendix Table A12 reports the measured reference, complete-target-roster, and matched-target covariate ranges.

Mean injected-minus-bare movement along the operational observed–bare axis is +0.023, with median +0.0175, in training-bare standardized-score units. Twenty-seven targets move toward the observed-positive side and 16 in the opposite direction; individual movements range from −0.453 to +0.998. The mean signed movement is therefore near zero despite heterogeneous pairwise shifts. The pooled injected-versus-bare AUC on this axis is 0.509. Unlike G_{domain}, this analysis asks how adding a fixed source moves the same target along an axis fitted between observed systems and bare targets. Mean movements in Stein, ImageNet, and pixel features are +0.235, +0.025, and −0.029, respectively. The Stein movement distribution is asymmetric: its median is −0.069, 12 movements are positive and

Table 5. Domain-ranking AUC on the same 43 observed and 6,336 injected rows. The first four rows are cross-fitted domain probes. "Fold range" gives their minimum and maximum fold AUC; folds contain only 8 or 9 observed systems. The fixed operational lens head is a non-cross-fitted baseline on the same roster and weights.

Readout	Dimension	Domain-ranking AUC	Fold range	Evaluation
Operational penultimate	1 280	0.991340	0.985–0.999	Cross-fitted domain probe
Stein SSL	2 048	0.958187	0.929–1.000	Cross-fitted domain probe
Generic ImageNet	1 280	0.978528	0.953–0.994	Cross-fitted domain probe
Pixel statistics	18	0.980745	0.938–0.997	Cross-fitted domain probe
Fixed operational lens head	–	0.980306	–	Not cross-fitted

31 negative, and the range is −0.982 to +5.204. It therefore does not indicate that most matched targets move toward the observed side. Table 6 reports the corresponding pooled AUCs.

Among 18 statistics, the z-band median has univariate AUC 0.888, and omitting it lowers multivariate pixel AUC from 0.981 to 0.932 in G_{domain}. Separately, within the 43 G_{match} triplets, its unweighted finite-cohort mean is 0.011448 in observed references, 0.005088 in bare targets, and 0.005094 after injection. This matched bare–injected comparison shows that the leading observed–injected offset is already present before source insertion. This pixel-level result does not establish which features drive the operational detector.

5.4 Conditional recovery and fixed-target brightness response

Conditional G_{rec} recovery is 3.45 per cent (95 per cent spatial-group interval, 3.10–3.83 per cent) under equal weighting of observing cells, injection targets, and Einstein radii. This is 1,710 detections among 49,500 scenes generated from 4,500 targets in 2,187 spatial groups. Two bare targets exceed the same threshold. Across target–radius comparisons, 22 are positive both before and after injection, 1,688 become positive, none becomes negative, and 47,790 remain negative. The injection therefore raises the positive-call rate by 3.41 percentage points across the specified grid.

The grid applies no rendered-arc visibility or S/N admission rule; its estimand therefore averages all designed scenes rather than a visibility-selected subset. Scene-weighted recovery is 8.45, 1.90, and 0.12 per cent in intrinsic source-magnitude bins $[23, 24)$, $[24, 25)$, and $[25, 26)$, respectively (Table A8). The final design also contains 99 cell–radius estimates: 500 targets at each radius in each of nine occupied PSF–depth cells. Appendix A3 reports that conditional response surface and its one empty PSF–depth combination.

Figure 3 shows the equal-cell, equal-target response by Einstein radius. Recovery is 0.044 per cent at 0.5 arcsec, reaches 6.22 per cent at 2.5 arcsec, and is 6.02 and 5.47 per cent at 2.75 and 3.0 arcsec. The simultaneous 95 per cent band has half-width 0.82 percentage points.

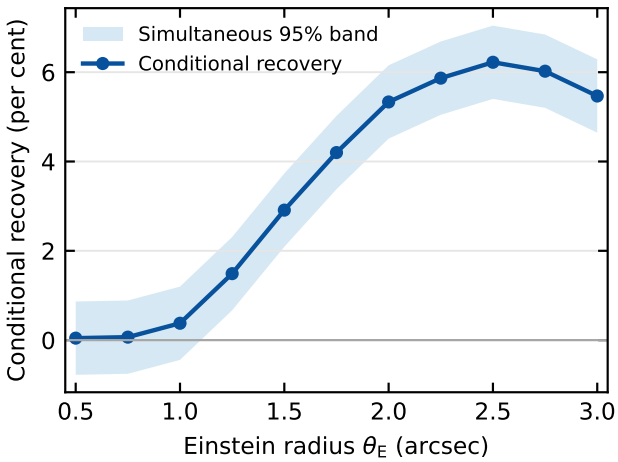


Figure 3. Conditional G_{rec} recovery at the $t_{10^{-3}}$ threshold versus Einstein radius. Points are equal-cell, equal-target estimates; shading is the simultaneous 95 per cent target-spatial-group multiplier band. The ordinate is conditional recovery over the designed scenes.

Alt text: A line rises from near-zero recovery at 0.5 arcsec to about 6.2 per cent at 2.5 arcsec, then declines modestly. A shaded simultaneous interval surrounds the line.

In the separate C_{bright} fixed-target experiment, expectation-image recovery is 6/43 and 7/43 at intrinsic magnitudes 18.5 and 19.5, peaks at 8/43 at magnitude 20.5, and then decreases to zero at magnitudes 24.5 and 25.5. Across all eight magnitudes, expectation-image recovery is non-monotonic; its peak at magnitude 20.5 exceeds each of the two brighter nodes by two and one targets, respectively. Photon-minus-expectation estimates range from −3.20 to +2.33 percentage points through magnitude 23.5, with finite-draw MCSE at most 1.07 points. These are responses of the same 43 targets with one retained source configuration per target and characterize detector response to source scaling over the tested magnitude range.

5.5 P_{noise} photon intervention

After equal weighting of cells, targets, and radii, modelled source-photon noise reduces recovery by 1.20 percentage points (95 per cent target-spatial-group interval, −1.39 to −1.03; MCSE, 0.014). Recovery declines from 3.45 per cent for expectation images to 2.25 per cent across photon draws.

By Einstein radius, the simultaneous band includes zero from 0.5 through 1.5 arcsec and lies below zero from 1.75 through 3.0 arcsec. The interval is conditional on the modelled source-photon law and frozen threshold; it excludes gain-model and threshold-calibration uncertainty. In particular, $g = 150\ \mathrm{e^-\ nmgy^{-1}}$ is the adopted approximate effective conversion for the source-photon intervention.

Table 6. G_{match} shared-axis pooled OOF AUC diagnostics for the 43 triplets. The axis is fitted on observed and bare rows; all entries are finite-cohort point estimates without formal intervals.

Representation	Obs.–bare	Obs.–inj.	Inj.–bare
Operational penultimate	0.998	0.998	0.509
Stein SSL	0.979	0.982	0.488
ImageNet	0.973	0.971	0.500
Pixel statistics	0.928	0.933	0.478

Appendix A9 reports the source-photon normalization replay, finite-gain diagnostic, and Monte Carlo errors.

5.6 P_{src} source-construction result

The P_{src} design contains 1,584 target–source pairs and eight photon draws per pair. Across the resulting 12,672 pair–draw rows per arm, the regularized GalSim-deconvolved structured F814W construction produces 220 detections (1.74 per cent), while its deterministic single-Sérsic twin produces 291 (2.30 per cent). At draw level, 83 discordant outcomes favour the Sérsic arm and 12 favour the structured arm.

The photon-averaged paired difference is +0.56 percentage points in favour of the single-Sérsic arm (95 per cent multiway target-and-source interval, +0.28 to +0.93; MCSE, 0.042). Within this fixed-intrinsic-flux construction, the result supports higher recovery for the single-Sérsic arm. The estimand compares the complete source-construction pipelines. Despite equal intrinsic band fluxes, the Sérsic arm has greater source-only lensed in-stamp r-band flux in 1,249 of 1,584 pairs and a greater source-only r-band peak in 1,516 pairs; the median Sérsic-to-structured ratios are 1.0417 and 1.1044, respectively. The higher lensed flux and peak intensity may contribute to the recovery difference, but the experiment does not isolate their individual effects. Deleting any one source or target spatial group does not reverse the sign of the result.

5.7 Detector-checkpoint sensitivity

The validation-selected checkpoints span all 12 training runs defined in Section 2.8. Every model-specific threshold yields exactly 20 exceedances among the same 20,002 calibration negatives. The checkpoint panel covers only thresholded detector responses; the feature-space results use the primary epoch-160 model. Achieved-FPR counts range from 9 to 21 among 12,512 evaluation negatives; 11 of the 12 spatial-group intervals contain 0.1 per cent. Selected-cohort response ranges from 35/43 to 39/43, and conditional G_{rec} from 1.28 to 3.15 per cent.

Across the validation-selected checkpoints, P_{noise} ranges from −1.16 to −0.51 percentage points, with all 12 intervals below zero. Every P_{src} estimate is positive, from +0.07 to +0.43 points, but only 7/12 threshold-conditional intervals exclude zero. Under the terminal-epoch policy all point-estimate directions are retained, all 12 P_{noise} intervals remain below zero, and 8/12 P_{src} intervals exclude zero. The directions are therefore consistent across the examined policies, while source-effect interval exclusion is policy-dependent. These are correlated variants of one training programme; Appendix A6 reports the full results and exploratory-policy qualification.

5.8 Renderer qualification

Across 142 realized scenes, production and reference renders agree within the reported numerical tolerances. The largest relative L_1 difference is 0.0077, the largest centroid difference 0.023 pixels, and the minimum pixel correlation 0.9999.

No G_{rec} or P_{src} scene uses the renderer's softened near-circular branch. Production-support SIE tests pass; the separate boundary-specific test does not meet its predefined criterion.

6 DISCUSSION

The operational detector's penultimate features strongly distinguish the selected observed and injected populations, showing that its trained representation can serve as a detector-specific diagnostic. The same separation appears in self-supervised survey features, generic ImageNet features, and elementary pixel statistics, so it is not confined to the lens-trained representation.

The controls add two distinct observations: a leading pixel-statistic offset predates insertion, and matched injections have a near-zero mean signed movement along the separately fitted operational observed–bare axis, with heterogeneous pairwise shifts. The separate recovery, source-photon, source-construction, and fixed-target brightness experiments characterize responses to controlled image changes. The matched controls locate differences that predate injection, while the recovery experiments measure how the fixed detector responds when source photons, source construction, or brightness are changed.

6.1 What the recovery measurements represent

The detector responds to 39/43 selected confirmed systems but to 3.45 per cent of the fixed injection grid. These summaries describe different cohorts and weighting schemes. The selected systems come from heterogeneous discovery programmes, whereas G_{rec} averages equally over nine observing cells, 4,500 targets, and 11 Einstein radii (Huang et al. 2020, 2021; Stein et al. 2022; Storfer et al. 2024; Sonnenfeld et al. 2023; Sonnenfeld 2024).

On the 43 fixed targets, expectation-image recovery peaks at magnitude 20.5 and is lower at the two brighter nodes. Thus recovery is non-monotonic over the tested intrinsic-magnitude range.

6.2 Feature diagnostics, matching, and mass-model scope

The fixed lens head is one affine projection of the operational feature vector. Its AUC of 0.980 shows that the existing score already expresses a strong separating direction. Using the same cross-fitted readout for all four representations shows whether the operational features are uniquely sensitive to population identity. The probe and fixed head are complementary readouts of the same feature vector, while the pixel-statistic AUC of 0.981 shows that elementary image summaries also carry a strong separating direction.

The leading native-pixel offset is already present between the observed references and bare targets (Section 5.3). Foreground light, background estimation, survey processing, and sample selection are all plausible contributors to that offset. The measured offset is present before insertion; these analyses do not separate the possible contributions listed above.

G_{match} asks how one fixed injection moves each matched target along an observed–bare axis. Its operational mean is near zero, but individual shifts range from −0.453 to +0.998. The near-zero cohort mean therefore coexists with substantial individual movements. Matching reduces differences in the five recorded central-object and observing quantities. The observed–bare readout describes differences between complete cutouts; insertion measures movement along that readout on the same targets. Redshift, velocity dispersion, stellar mass, source properties, and environment remain outside that matching set. The radius curve similarly describes response over assigned grid values rather than target-specific inferred lens models.

6.3 Arc-photon-noise treatment

The negative aggregate photon effect persists across the 12 validation-selected checkpoints (Section 5.7). The primary checkpoint's radius-specific band is a different summary: it includes zero through 1.5 arcsec and lies below zero at larger tested radii (Section 5.5).

Only the extra shot-noise term associated with injected flux is changed. The observed target already contains its realised sky, detector, coaddition, neighbours, and foreground light. The adopted effective gain is approximate, and the intervention operates on post-coadd cutouts rather than reconstructing exposure-level correlated noise or resampling. Eight draws control finite-photon Monte Carlo error for the adopted $g = 150\,\mathrm{e}^{-}\,\mathrm{nmgy}^{-1}$ source-photon law. Gain-model and threshold-calibration uncertainty remain outside the reported interval.

6.4 Source-construction result

Under the tested construction, single-Sérsic recovery is about 0.56 percentage points higher than structured-source recovery. All 12 validation-selected checkpoint estimates have the same sign with 7/12 intervals excluding zero. Under the terminal-epoch policy all 12 estimates retain that sign, while 8/12 intervals exclude zero. The direction is therefore consistent across the examined policies, but interval exclusion is checkpoint-policy-dependent. The estimand compares the two complete source-construction pipelines.

The empirical arm is constructed from selected high-S/N F814W images by deconvolution and reconvolution to a common resolution, followed by non-negative projection, centring, support restriction, and normalization. Its spatial map therefore contains both source structure and the effects of measurement and regularization. Each arm reuses its own F814W-derived map across g, r, z; the tested construction has no colour gradients or chromatic morphology. Both arms share the selected targets, assigned SIE-plus-shear mappings, effective PSF, modelled photon law, threshold, and post-coadd insertion. In the noiseless known-source test, global flux, centring, and normalization errors remain within the stated tolerances (Appendix A14).

Future paired studies can vary source support, regularization, correlated measurement noise, and band-dependent morphology while continuing to hold targets, lens mappings, fluxes, and photon draws fixed.

6.5 Renderer qualification and numerical scope

Agreement in the fixed 142-scene validation cohort supports the numerical implementation within its tested configurations (Appendix A15). These numerical checks do not test the astrophysical adequacy of the adopted source, lens, PSF, or detector models.

6.6 Relation to published pipelines and resolution

Published pipelines differ in their source constructions, deflector information, and observing regimes: fully simulated *Euclid*-like images expose selection dependence on source and lens parameters (Herle et al. 2024), whereas HSC and other recent pipelines use real galaxy stamps and, in some cases, more tightly constrained deflector information (Cañameras et al. 2021; Rojas et al. 2022; Schuldt et al. 2023; Cañameras et al. 2024). These approaches vary choices beyond the paired structured–Sérsic intervention studied here.

Resolution is especially important. At DR10 resolution, the effective PSF smooths the rendered lensed images and reduces the observable fine structure inherited from the source templates. HST, *Euclid*, and seeing-limited ground-based data resolve different source structure, so the magnitude and direction of source-construction effects can vary with resolution. Source, mass, PSF, detector, and processing choices should therefore be tested at the resolution at which a selection function will be applied.

6.7 Limitations and future work

Inference is restricted to the sampled systems and tested pipelines. Generalizing these results to another lens population is limited by missing physical covariates, mass parameters assigned independently of the target galaxies, single-band source morphology, post-coadd insertion, approximate PSF and gain models, and unknown overlap with Stein's pretraining data. The operational evaluations retain the detector's trained 20–32-pixel annulus and clip-10 map, making that mapping part of the operational estimand.

The detector panel contains 12 correlated variants of one architecture and training pipeline. Sampler and schedule changes are evaluated only at seed 42, the positive-example-set contrast changes several properties together, and no between-checkpoint contrast interval is attached. Each estimate conditions on an empirically calibrated threshold, so the panel combines training variation with calibration-tail variation. It therefore tests variation within this training family rather than across independent detector architectures. The operational feature analysis uses only the primary checkpoint. All four probes share the 43 observed references; dependency-closed folds keep linked images together, and component-equal weights prevent repeated injections from receiving disproportionate weight.

Future work should first construct a deflector sample with reliable redshifts, velocity dispersions or other mass proxies, then assign mass and environment jointly with observed light. Controlled mass-model factors should include density slope, composite baryonic and dark components, external convergence, multiple deflectors, and line-of-sight structure. Each factor should be varied within matched targets rather than inferred from an unpaired population contrast.

Source tests should use multiple real-galaxy catalogues and latent-source methods, band-dependent morphology and colour gradients, with support, regularization, and correlated noise varied explicitly. The structured–Sérsic intervention should be repeated at space-based and ground-based resolution under the same underlying systems. Exposure-level injection before coaddition would test PSF, noise

correlation, deblending, masking, and processing provenance that cutout-level addition leaves fixed.

Future studies should predefine the simulated population, detector, and thresholds; preserve system- or target-level grouping; report support in every conditional cell; and combine unpaired comparisons with bare-target and paired controls. Repeating the diagnostic under such controlled pipeline changes can identify which changes alter observed–injected separation. Pairing those changes with selection-function and downstream population-inference metrics would connect the diagnostic to survey performance.

7 CONCLUSIONS

We demonstrated a representation-level diagnostic of observed–injected differences in selected DESI Legacy Imaging Surveys DR10 images. Linear probes reuse frozen trained features to evaluate this separation directly. Strong held-out discrimination in the operational finder's penultimate representation is accompanied by discrimination in survey self-supervised features, generic ImageNet features, and elementary pixel statistics under the common dependency-aware procedure. The separation appears in the operational detector, two external learned representations, and simple pixel statistics.

A leading pixel-statistic offset largely predates insertion, and the separately fitted operational observed–bare axis has a near-zero mean signed movement with heterogeneous individual shifts.

Separately, the finder recovers 3.45 per cent of the full grid without a visibility cut. Modelled source-photon sampling lowers this response, and the tested paired single-Sérsic construction has higher recovery than its regularized structured counterpart. The brightness control shows how the same targets respond to source scaling, while the checkpoint panel shows which intervention results persist across trained models. The combined results show why future injection pipelines should be compared on matched foreground populations and observing conditions, with bare-target controls used to separate pre-existing image differences from the effects of source insertion.

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This work used PyTorch and torchvision for neural-network training and inference; NumPy, SciPy, pandas, and pyarrow for

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DATA AVAILABILITY

DESI Legacy Imaging Surveys DR10 imaging and catalogues are publicly available at https://www.legacysurvey.org/dr10/. The GalSim COSMOS 25.2 source products are the archived dataset of Mandelbaum et al. (2012), https://doi.org/10.5281/zenodo.3242143. The configurations, analysis code, compact score and feature tables, out-of-fold predictions, and provenance records used to inspect the reported calculations are available at https://github.com/aditrivaishnavi/stronglens-realism-gap. A public, versioned archival release of the remaining redistributable project products will be deposited no later than publication, and its DOI will be added to the final article. It will include the exact primary EfficientNetV2-S checkpoint (SHA-256 `5d1082839aac7a0bfb89342f84a9a533b22e1111c15cb4711255ed1b524a2b8d`), the 12-checkpoint panel manifest and hashes, and the compact panel score and result tables. If repository size limits require it, the checkpoint weights will occupy separate DOI-linked records. Because of their size, the generated image banks will be archived separately from the compact analysis products.

Third-party DR10 and COSMOS pixels remain subject to their source archives and licences rather than being redistributed as original data by this work.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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APPENDIX A: TECHNICAL VALIDATION OF THE REPORTED STUDY

A1 Frozen detector, training record, and scoring contract

All primary detector-response analyses use the same EfficientNetV2-S checkpoint: training seed 42, epoch 160, SHA-256 `5d1082839aac7a0bfb89342f84a9a533b22e1111c15cb4711255ed1b524a2b8d`. Section 2.7 defines the primary training and checkpoint rules; Section 2.8 separately defines the 12-run sensitiv-

ity panel. The archived training configuration and checkpoint hash identify the exact primary model.

The retained checkpoint-validation summary contains 61,437 negatives and 778 noisy catalogue-positive labels. At native-logit precision its ROC AUC is 0.96531. Average precision is 0.79659 from native logits and 0.79173 from float32 sigmoid scores because 287 probabilities saturate at one; the float64-probability Brier score is 0.005169. These values describe the recorded noisy-label validation set. The operating threshold and selected-system evaluation use their dedicated cohorts.

Applying Equation 8 to the 20,002 calibration rows gives $t = 1.5717214345932007$ (float32 bits 1070149163), with 20 exceedances and comparator $\ell \geq t$. Scoring uses the checkpoint above in evaluation mode, a fixed batch width of 64, native float32 logits, stored batch-normalisation statistics, inactive dropout, and an MPS (*Metal Performance Shaders*) backend whose software and hardware profile is checksum-bound. Padding repeats the final real row; filler outputs are discarded. The bitwise checks apply to this MPS backend and batch configuration; other hardware or software environments may produce small numerical differences.

The separate achieved-FPR endpoint contains 12,512 catalogue-negative rows in 1,089 spatial groups and was not used to recalibrate t. It has 13 exceedances, or 0.001039. The row-Wilson interval is [0.000607, 0.001777], and the spatial-group multiplier interval is [0.000546, 0.001656]. A separate author-run implementation checks the manifest and 12,512 payload arrays and reproduces the native score bits, threshold decisions, and result calculation.

A2 Four-representation probe diagnostics

The operational feature extraction uses the primary checkpoint's 1,280-component pooled penultimate output. The final G_{domain} feature matrices contain 6,379 rows, and the final domain and matched manifests together contain 6,444 unique pixel payloads. On the recorded MPS backend, applying the retained linear classifier to the final operational features reconstructs all 6,379 native logits bit for bit. Separate same-version refits from each of the four persisted final domain matrices reproduce every OOF score exactly. These checks verify the archived arrays and estimator implementation; they are not independent scientific replications.

Table A2 gives the five fold diagnostics. Each fold contains only 8 or 9 observed systems. The pooled OOF AUC is the prespecified finite-cohort summary; the fold values show the support and variation across its five fitted readouts.

The fixed operational head uses the same component/class weights but no domain-probe fit. Its component-weighted AUC is 0.9803064184, compared with row-weighted AUC 0.9812331160. Median native logits are 19.6848 for observed references and −22.6331 for injections. These are selected-population descriptives, not calibrated probabilities.

A3 Conditional PSF–depth–radius response

All 99 occupied cell–radius summaries were reproduced directly from the final 49,500 score rows. Each entry contains 500 distinct targets. The design has five PSF nodes and two depth nodes but occupies nine of the ten possible combinations; PSF 1.05 arcsec at depth 24.0 mag is empty. Figure A1 shows the occupied support rather than interpolating the missing cell.

A4 Fixed-target brightness and source-photon response

The brightness control contains 344 target–magnitude cells, 344 expectation images, and 2,752 photon realizations. Table A4 reports equal-target recovery; photon rates first average eight draws within target. Eight-draw finite-photon MCSE is at most 1.071 percentage points. Zero MCSE at the faint nodes reflects eight all-negative draw-specific cohort rates, not a general precision guarantee.

This finite-cohort detector-response test characterizes the fixed targets and tested source scalings; intrinsic magnitude supplies the controlled axis.

Table A1. Full-precision primary results underlying the rounded main-text values. Intervals are the procedures defined in Section 4; movements are fold-standardised units and effects are percentage points.

Quantity	Estimate	Interval or diagnostic
Threshold	1.5717214345932007	float32 bits 1070149163
Achieved FPR	13/12 512 = 0.103900%	spatial-group 0.054580–0.165564%
Selected systems	39/43 = 90.697674%	finite cohort
G_{rec}	3.454545%	3.097551–3.830632%
G_{domain} AUC	0.991340 / 0.958187 / 0.978528 / 0.980745	operational / Stein / ImageNet / pixel
Fixed-head domain ranking	0.980306	existing operational lens logit
G_{match} movement	+0.023183 / +0.235027 / +0.024792 / -0.029207	operational / Stein / ImageNet / pixel
P_{noise}	-1.203030	[-1.387747, -1.027768]; MCSE 0.013886
P_{src}	+0.560290	[+0.283912, +0.932231]; MCSE 0.042076

Table A2. G_{domain} weighted OOF AUC by frozen dependency fold. Observed support is 8, 9, 8, 9, and 9 systems in folds 0–4; simulated support is 1,276, 1,265, 1,265, 1,265, and 1,265 rows.

Feature protocol	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
Operational penultimate	0.985194	0.996212	0.992515	0.994124	0.999320
Stein SSL	0.952142	0.974942	0.999945	0.976933	0.929293
Generic ImageNet	0.969132	0.953089	0.993990	0.978681	0.992036
Pixel statistics	0.937664	0.996503	0.993990	0.974699	0.986694

Table A3. Native operational-logit distribution on the exact G_{domain} populations. Entries are finite-cohort quantiles.

Population	5%	25%	50%	75%	95%
Observed	-1.70	13.58	19.68	23.54	26.08
Injected	-27.07	-24.32	-22.63	-19.73	0.20

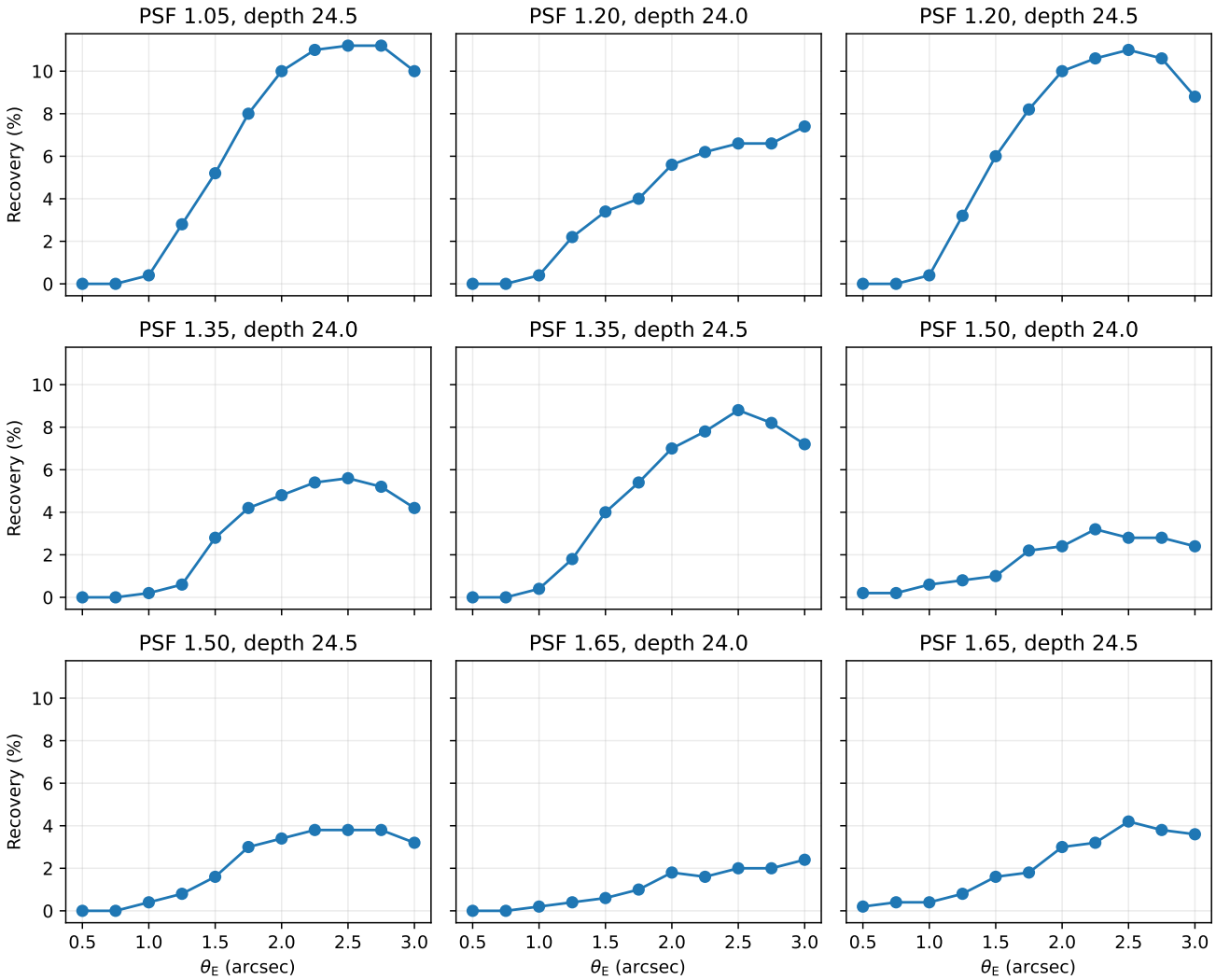


Figure A1. Conditional G_{rec} recovery versus Einstein radius in each occupied PSF–depth cell. Every point contains 500 targets. The panels are descriptive support strata covering nine of the ten possible PSF–depth combinations.

Alt text: Nine panels show recovery curves over eleven Einstein radii for the occupied combinations of five PSF values and two depth values. The PSF 1.05, depth 24.0 combination is absent from the design.

Table A4. Fixed-target C_{bright} recovery percentages under the trained preprocessing and common threshold.

$m_{r,\mathrm{src}}$	Expectation	Photon
18.5	13.953	12.500
19.5	16.279	13.081
20.5	18.605	18.314
21.5	11.628	13.953
22.5	9.302	7.558
23.5	2.326	1.744
24.5	0.000	0.000
25.5	0.000	0.000

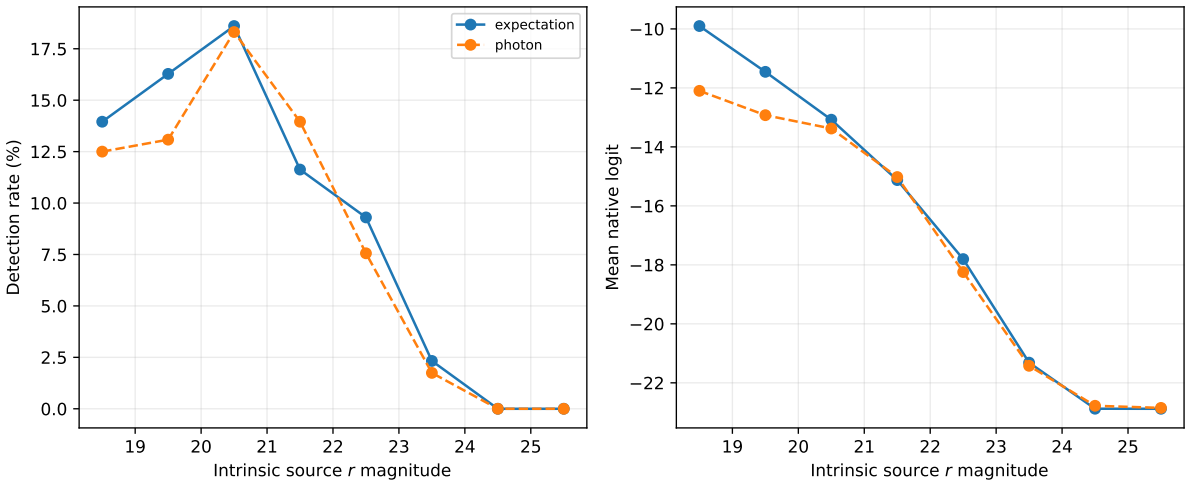


Figure A2. Detection fraction and mean native logit versus intrinsic source magnitude for the fixed 43-target brightness experiment. Expectation and photon-realized images use the trained preprocessing and common threshold. The detection fraction peaks near magnitude 20.5, whereas the mean logit declines across the tested magnitude range. These are finite-cohort response summaries on 43 fixed targets.

Alt text: Two panels show expectation and photon-realized images over intrinsic source magnitudes 18.5 to 25.5. Detection fractions reach their maximum at 20.5 and are zero at 24.5 and 25.5. Mean logits decline toward fainter magnitudes. The photon and expectation curves are close but not identical.

A5 P_{noise} draw-count pilot

The score-blind pilot contains 288 targets and 3,168 scenes, with 64 draws per scene. MCSE and the target-sampling standard error are on the probability scale; the latter is 0.003553. A candidate passes when MCSE is at most 0.002 and at most 20 per cent of the target-sampling standard error. One draw is diagnostic only because its MCSE is unavailable; eight is the smallest count satisfying both criteria.

Table A5. P_{noise} design-pilot photon-draw selection.

Draws	MCSE	MCSE/SE	Absolute criterion	Relative criterion
1	unavailable	unavailable	No	No
2	0.001119	0.315	Yes	No
4	0.000791	0.223	Yes	No
8	**0.000559**	**0.157**	**Yes**	**Yes**
16	0.000396	0.111	Yes	Yes
32	0.000280	0.079	Yes	Yes
64	0.000198	0.056	Yes	Yes

A6 Detector-checkpoint sensitivity

Tables A6 and A7 report the runs defined in Section 2.8, selected and calibrated as in Section 4.2. The within-run selection metric is average precision from float32 sigmoid outputs on that run's noisy-label validation population; an exact tie retains the earliest epoch. The primary detector remains the terminal epoch-160 checkpoint.

The two-policy comparison, including designation of the validation-selected policy for the main presentation, is exploratory. Under both policies all 12 aggregate P_{noise} estimates are negative with intervals below zero, and all 12 P_{src} estimates are positive. Source-effect intervals exclude zero for 7/12 validation-selected checkpoints and 8/12 terminal-epoch checkpoints. These related analyses describe policy sensitivity within the same training runs.

A separate author-run verification implementation reproduces the validation-selected epochs, recorded checkpoint hashes, thresholds, row identities, estimates, intervals, and MCSE values.

A7 Descriptive attribution of the pixel-statistic probe

Using the same dependency-closed folds, component weights, and fold-local standardisation, a secondary descriptive suite refits each of the 18 statistics alone and after its omission, and likewise evaluates each photometric band. Median z-band pixel intensity is strongest: its univariate OOF AUC is 0.888, omitting it reduces the multivariate AUC from 0.981 to 0.932, and its coefficient is positive in all five folds. A z-band-only probe gives AUC 0.968 and omitting that band gives 0.929. Here z denotes the band, not redshift. Median intensity can encode foreground surface brightness, background level, or sample selection, so the attribution summarizes a mixture of image-population properties. All 43 fits were separately recomputed by the authors as finite-cohort diagnostics.

Table A6. Final checkpoint-sensitivity endpoints. Each t_m yields exactly 20 threshold exceedances among the same 20,002 calibration negatives. Achieved-FPR counts use the same 12,512-row evaluation cohort; their percentages and the G_{rec} percentages are followed by 95 per cent spatial-group intervals. Selected-system entries are exact counts among the same 43 systems. Baseline denotes natural-prevalence sampling with cosine scheduling; the other variant labels name the one changed training factor. "Both sources" uses reviewed and additional catalogue candidates; "Reviewed only" omits the additional candidates. Validation AP is the within-run checkpoint-selection value; values for the two positive-example compositions are computed on validation populations with 778 and 64 noisy positive labels, respectively, and should not be compared as if their labels or prevalence were the same. The target FPR is an empirical calibration rate rather than a guaranteed population rate.

Variant	Positive examples	Seed	Epoch	Val. AP	t_m	Achieved FPR: count; per cent [interval]	Selected	G_{rec}: per cent [interval]
Baseline	Both sources	42	19	0.8104	1.153126	11; 0.0879 [0.0434, 0.1454]	35/43	2.5717 [2.2801, 2.8757]
Baseline	Both sources	123	17	0.8181	0.621170	13; 0.1039 [0.0550, 0.1656]	37/43	2.4263 [2.1344, 2.7286]
Baseline	Both sources	2026	93	0.8053	3.659930	16; 0.1279 [0.0738, 0.1948]	37/43	3.1535 [2.8163, 3.5084]
Baseline	Reviewed only	42	103	0.7764	-9.486459	17; 0.1359 [0.0773, 0.2093]	37/43	1.8485 [1.6015, 2.1177]
Baseline	Reviewed only	123	93	0.7704	-8.606620	21; 0.1678 [0.1047, 0.2454]	36/43	1.4525 [1.2305, 1.6973]
Baseline	Reviewed only	2026	43	0.7679	-1.618260	16; 0.1279 [0.0700, 0.2059]	36/43	1.2808 [1.0801, 1.4987]
50:50 sampler	Both sources	42	62	0.7922	6.281433	12; 0.0959 [0.0502, 0.1552]	35/43	2.8081 [2.4995, 3.1276]
50:50 sampler	Reviewed only	42	37	0.7704	-7.143696	17; 0.1359 [0.0751, 0.2150]	37/43	1.3778 [1.1665, 1.6091]
Step schedule	Both sources	42	18	0.8109	1.875446	9; 0.0719 [0.0333, 0.1255]	35/43	2.1071 [1.8303, 2.3978]
Step schedule	Reviewed only	42	87	0.7653	-2.322129	14; 0.1119 [0.0620, 0.1777]	35/43	1.3636 [1.1628, 1.5833]
Additional seed	Reviewed only	7	86	0.7516	-7.043954	14; 0.1119 [0.0611, 0.1763]	39/43	1.3980 [1.1897, 1.6259]
Additional seed	Reviewed only	2027	99	0.7695	-9.008804	19; 0.1519 [0.0872, 0.2345]	36/43	1.4040 [1.1922, 1.6399]

Table A7. Final checkpoint sensitivity of the paired interventions. P_{noise} is modelled-photon-minus-expectation and P_{src} is Sérsic-minus-structured. Every row uses its checkpoint-specific threshold yielding exactly 20 exceedances among 20,002 calibration negatives. Estimates, intervals, and MCSE values are percentage points. The 95 per cent intervals use one global target-spatial-group multiplier for P_{noise}. For P_{src}, each pair receives the product of global target-spatial and source multipliers, with source multipliers normalised within the 16 source strata. MCSE uses the eight photon draws. Baseline denotes natural-prevalence sampling with cosine scheduling; the other variant labels name the one changed training factor. "Both sources" uses reviewed and additional catalogue candidates; "Reviewed only" omits the additional candidates. Intervals condition on the displayed checkpoint-specific thresholds and omit calibration-sample variation.

Variant	Positive examples	Seed	Epoch	P_{noise} [95% interval]	MCSE	P_{src} [95% interval]	MCSE
Baseline	Both sources	42	19	-0.9679 [-1.1321, -0.8137]	0.0122	+0.3946 [+0.1622, +0.7152]	0.0569
Baseline	Both sources	123	17	-0.8379 [-0.9854, -0.7017]	0.0120	+0.2683 [+0.0278, +0.5723]	0.0207
Baseline	Both sources	2026	93	-1.1644 [-1.3399, -0.9983]	0.0130	+0.4261 [+0.1349, +0.8361]	0.0651
Baseline	Reviewed only	42	103	-0.9672 [-1.1183, -0.8282]	0.0088	+0.1184 [-0.0456, +0.3366]	0.0303
Baseline	Reviewed only	123	93	-0.7341 [-0.8719, -0.6123]	0.0073	+0.0947 [-0.0411, +0.2952]	0.0442
Baseline	Reviewed only	2026	43	-0.7813 [-0.9180, -0.6529]	0.0067	+0.0710 [-0.1679, +0.3166]	0.0251
50:50 sampler	Both sources	42	62	-0.9982 [-1.1705, -0.8406]	0.0129	+0.3946 [+0.1512, +0.7127]	0.0752
50:50 sampler	Reviewed only	42	37	-0.5111 [-0.6272, -0.4054]	0.0091	+0.2446 [+0.0809, +0.5178]	0.0469
Step schedule	Both sources	42	18	-0.7162 [-0.8697, -0.5764]	0.0114	+0.3157 [+0.1297, +0.6001]	0.0506
Step schedule	Reviewed only	42	87	-0.8048 [-0.9429, -0.6789]	0.0072	+0.1736 [+0.0530, +0.4061]	0.0231
Additional seed	Reviewed only	7	86	-0.8535 [-1.0025, -0.7184]	0.0073	+0.0947 [-0.0406, +0.2676]	0.0207
Additional seed	Reviewed only	2027	99	-0.7856 [-0.9292, -0.6546]	0.0077	+0.0947 [-0.0740, +0.2794]	0.0267

A8 G_{rec} brightness stratifications

Table A8 partitions the 49,500-scene grid and 1,710 detections by intrinsic source magnitude and by post-hoc source-only in-stamp r-flux quartiles. Intrinsic magnitude is shared over the eleven radii of a target, while one target can enter several rendered-flux quartiles. Rates retain the original scene weights within bin. The bins are descriptive intrinsic-magnitude and rendered-flux strata.

Table A8. Descriptive G_{rec} brightness stratifications at the primary threshold.

Stratification	Bin	Detections / scenes	Recovery (%)
Intrinsic $m_{r,\mathrm{src}}$	$[23, 24)$	1,367 / 16,170	8.454
Intrinsic $m_{r,\mathrm{src}}$	$[24, 25)$	323 / 16,995	1.901
Intrinsic $m_{r,\mathrm{src}}$	$[25, 26)$	20 / 16,335	0.122
Rendered source r-flux	Lowest quartile	11 / 12,375	0.089
Rendered source r-flux	Second quartile	7 / 12,375	0.057
Rendered source r-flux	Third quartile	241 / 12,375	1.947
Rendered source r-flux	Highest quartile	1,451 / 12,375	11.725

A9 Photon replay, normalization, and gain sensitivity

The replay cohort contains 4,568 registered photon-draw realizations derived from 571 deterministic expectation images and includes both score-blind coverage and deliberately selected crossing cases. Replayed native score bits agree with the retained scores for these realizations. Using each deterministic expectation image's annular normalization statistics instead of recalculating them for its photon-draw realization changes three decisions. Most selected threshold changes therefore persist under expectation-anchored normalization. This replay shows normalization sensitivity in the inspected cohort, but it does not cover the complete photon bank.

The separate gain diagnostic contains 198 scenes from 18 targets in 17 spatial groups. Deterministic recovery is 5/198, or 2.5253 per cent. Sixteen keyed draws per scene give Table A9. The registered eight gain-150 draws on these same scenes give 39/1,584, or 2.4621 per cent, compared with 2.4937 per cent in the 16-draw gain-150 sample. This agreement on common scene support shows that the gain comparison applies only to these 198 scenes and should not be compared directly with the full-grid estimate.

Table A9. Fixed-cohort gain sensitivity. Changes are relative to deterministic expectation. Gain is in $\mathrm{e}^{-}\ \mathrm{nmgy}^{-1}$; change and MCSE are percentage points. MCSE is conditional photon Monte Carlo error, not a target-population interval.

Gain	Positive / 3,168	Recovery (%)	Change / MCSE
75	59	1.8624	−0.6629 / 0.1961
150	79	2.4937	−0.0316 / 0.1813
300	82	2.5884	+0.0631 / 0.1299
600	86	2.7146	+0.1894 / 0.1044

A10 Author-run endpoint and experiment checks

Table A10 distinguishes checks of the external and operational feature protocols from checks of the thresholded experiments. The checks separately verify feature extraction, classifier-head reconstruction, out-of-fold calculations, and estimator refitting from the archived arrays. Numerical agreement is evaluated under the recorded software versions and backend.

Exactly 39/43 selected systems exceed the primary threshold. A separate author-run implementation rehashes the 43 payload arrays and reproduces the native score bits, decisions, and result. The endpoint is the response of this finite cohort; no population interval is attached.

Figure A3 shows all four non-detections rather than a selected subset, under one common display mapping.

Per-scene convergence diagnostics are absent from the final manifest for 43,780 stored G_{rec} deterministic expectation images. For 2,364 of those retained rows, the diagnostics remain in their bound generating manifest; 41,416 rows lack diagnostics in both supplied manifests. Numerical qualification therefore also uses the specified 142-scene reconstruction audit. One scene required order-48 quadrature; its identity and image checksum are recorded. For rows without diagnostics, the reconstructed manifest order field is not treated as a per-scene convergence record.

Table A10. Author-run checks of the reported experiment products. OOF denotes out-of-fold.

Analysis	Verified support	Author-run check
G_{rec}	49,500 rows; 4,500 targets; 2,187 spatial groups	MPS logits and score bindings were reproduced; both 10,000-draw multiplier calculations agreed.
G_{domain}, external/elementary	6,379 rows; 561 dependency components	Same-version refits from each persisted final matrix reproduce every OOF score exactly under the recorded folds, weights, scaler, and estimator.
G_{domain}, operational	6,379 rows; 1,280 features per row	The retained head reconstructs all 6,379 native logits bit for bit on the recorded MPS backend, and the same-version OOF refit is exact.
G_{match}, external/elementary	129 rows; 43 triplets; 43 dependency components	Current row identities, dependency closure, pair movements, and shared-axis AUC arithmetic were recomputed from the persisted final axis scores. This check recomputes the statistics from archived axis scores but does not repeat feature extraction.
G_{match}, operational	129 rows; 43 triplets; 43 dependency components	Pair movements and shared-axis AUC arithmetic were recomputed from persisted final axis scores; this is an arithmetic check rather than an independent feature extraction.
P_{noise}	49,500 deterministic expectations; 396,000 photon-draw realizations	The complete stored score table, primary point estimate, interval, and finite-draw MCSE were recomputed. The separate 4,568-row replay in Appendix A9 is a fixed implementation diagnostic.
P_{src}	1,584 pairs; 12,672 pair-draw rows	All 1,584 photon artefacts were rehashed; native score bits and decisions, pair averages, point estimate, interval, and finite-draw MCSE were separately recomputed.

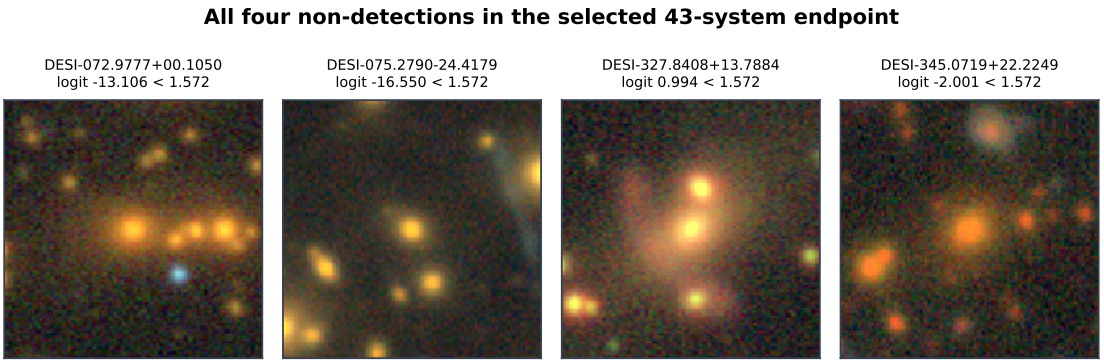


Figure A3. All four non-detections among the 43 selected confirmed systems at the primary threshold $t_{10^{-3}} = 1.5717214346$. Titles give the catalogue name and native-float32 logit. Every panel uses the same complete 101×101-pixel Legacy Survey $z/r/g$ display mapping defined in Equation 16; no panel-specific intensity rescaling or crop is applied. The figure shows all four missed systems, not a representative sample of the 43-system cohort.

Alt text: Four square Legacy Survey colour cutouts arranged in one row. Each title identifies a confirmed system and shows its logit below the common threshold. The images contain bright foreground objects and surrounding structures with varied colour and configuration.

A11 Object-level provenance of the 43-system endpoint

The selected endpoint is not a uniformly dual-spectroscopic sample. Table A11 lists the object identifier, confirmation route, principal object-level publication, and one representative recorded foreground and source redshift for every evaluated system. Multiple measurements and the complete evidence record are provided in `confirmed_43_provenance.csv` in the reproducibility materials; "–" means that no representative role-assigned redshift is recorded. The confirmation-route column identifies whether displayed values are secure, probable, low-confidence, or unverified.

Table A11. Object-level provenance of the selected 43-system endpoint.

Identifier	Confirmation route	Principal evidence	$z_{\rm foreground}$	$z_{\rm source}$
0361p000_6437	secure foreground and source spectra	(Gladders, Yee & Ellingson 2002; Gladders et al. 2003)	0.773	4.8786
0442m235_7484	secure foreground and source spectra	(Zitrin et al. 2011; Nakane et al. 2025)	0.5049	1.09
0537m132_1571	secure foreground and source spectra	(Lin et al. 2026)	0.3866	2.30117
0572m102_2469	secure foreground and source spectra	(Lin et al. 2026)	0.74493	1.60376
0656m280_3774	secure foreground and source spectra	(Lin et al. 2026)	0.62107	1.17482
0728p000_6851	secure source; foreground spectrum unverified	(Sand et al. 2005)	0.430	2.007
0734m102_4599	secure foreground and source spectra	(Richard et al. 2010; Lin et al. 2026)	0.24820	1.043
0750m307_2398	secure foreground and source spectra	(Lin et al. 2026)	0.44137	1.44884
0753m245_3999	secure foreground and source spectra	(Lin et al. 2026)	0.32059	2.8290
0862m265_8849	secure foreground and source spectra	(Lin et al. 2026)	0.27534	2.17441
0870m362_8874	secure foreground and source spectra	(Lin et al. 2026)	0.3011	0.84611
0910m360_2444	secure foreground and source spectra	(Lin et al. 2026)	0.4883	1.43211
1219p105_7293	secure foreground and source spectra	(Huang et al. 2026; Lin et al. 2026; Tran et al. 2022)	0.475	1.23742
1409p185_769	secure foreground and source spectra	(Huang et al. 2026; Tran et al. 2022)	0.873	2.41673
1434p092_2623	secure lens; probable source spectrum	(Tran et al. 2022)	0.743	2.43243
1601p187_3957	secure foreground and source spectra	(Lin et al. 2026; Tran et al. 2022)	0.314	0.87872
1614m087_3206	secure lens; low-confidence source spectrum	(Lin et al. 2026)	0.82754	2.079
1655m060_944	secure foreground and source spectra	(Zaborowski et al. 2023; Huang et al. 2026)	0.4834	1.67511
1658p122_150	imaging/model; no secure dual spectra	(Huang et al. 2026)	–	–
1671p042_499	secure foreground and source spectra	(Lin et al. 2026)	0.35153	1.28023
1683p240_1137	secure foreground and source spectra	(Kubo et al. 2009)	0.3361	0.766
1686p167_5461	secure foreground and source spectra	(Stark et al. 2013; Lin et al. 2026)	0.537	1.718
1744p147_5445	secure foreground and source spectra	(Lin et al. 2026)	0.56486	2.41864
1894p150_8726	secure foreground and source spectra	(Huang et al. 2026; Tran et al. 2022)	0.57160	1.16149
1908p212_1676	secure foreground and source spectra	(Lin et al. 2026)	0.34832	1.48152
1928p017_9908	imaging/model; no secure dual spectra	(Huang et al. 2026)	–	–
1976p147_245	secure foreground and source spectra	(Lin et al. 2026)	0.26144	0.85834
2006p037_9179	secure foreground and source spectra	(Lin et al. 2026; Tran et al. 2022)	0.353	1.01590
2026p047_4274	secure foreground and source spectra	(Huang et al. 2026; Lin et al. 2026; Tran et al. 2022)	0.336	1.16757
2168m067_3815	secure foreground and source spectra	(Zaborowski et al. 2023; Tran et al. 2022)	0.26500	1.51409
2204p147_3771	secure foreground and source spectra	(Stark et al. 2013; Pettini et al. 2010)	0.741	1.433
2276p205_1624	imaging/model; no secure dual spectra	(Huang et al. 2026)	–	–
2349p167_2044	secure foreground and source spectra	(Huang et al. 2026; Tran et al. 2022; Barone et al. 2026)	0.409	1.833
2530p280_351	imaging/model; no secure dual spectra	(Huang et al. 2026)	–	–
2608p237_3728	secure foreground and source spectra	(Lin et al. 2026)	0.2296	0.97503
3108m062_2639	secure foreground and source spectra	(Huang et al. 2026; Tran et al. 2022; Barone et al. 2026)	0.79261	1.892
3277p137_6699	secure lens; probable source spectrum	(Huang et al. 2026; Tran et al. 2022; Barone et al. 2026)	0.20643	0.890
3409p280_5325	secure foreground and source spectra	(Lin et al. 2026; Tran et al. 2022)	0.343	0.96034
3418m020_1357	secure foreground and source spectra	(Lin et al. 2026; Sand et al. 2005)	0.330	2.237
3428m035_4814	secure foreground and source spectra	(Lin et al. 2026)	0.24592	1.05956
3431m042_1867	secure lens; low-confidence source spectrum	(Lin et al. 2026)	0.43519	0.58
3450p222_3227	secure foreground and source spectra	(Wen, Han & Jiang 2011; Stark et al. 2013)	0.443	1.93
3459p235_481	secure foreground and source spectra	(Lin et al. 2026)	0.27653	1.29593

A12 Injection-target and matching support

The 4,500 $G_{\rm rec}$ targets are catalogue-selected injection targets, not confirmed deflectors. Their Tractor types are 2,853 SER, 1,213 DEV, and 434 REX. Observed z magnitude has median 19.40, interquartile range 18.71–19.77, and range 13.29–20.00. The selection manifest contains no homogeneous catalogue photo-z field for these 4,500 rows. Photo-z, velocity dispersion, stellar mass, and environment therefore neither select these targets nor assign their SIE parameters.

Under the score-blind assignment in Section 4.5, all 43 references receive unique targets. There are 168–4,229 admissible targets per reference, and selected robust-IQR distances range from 0.108 to 1.140. Table A12 reports the measured support before and after matching.

Direct checks of the stored bare and injected arrays verified their pairing and the recorded target, source, lens, PSF, rendering, random, and flux identities. No target-total flux floor, post-lensing flux rescaling, detector score, or probe outcome enters matching or pair construction. The matching set covers five measured image and catalogue quantities; homogeneous deflector redshift, mass, environment, and observed-source brightness are unavailable across all references.

Table A12. Measured-covariate support for the 43 references, complete $G_{\rm rec}$ target roster, and 43 matched targets. Entries give minimum/median/maximum; redshift, mass, environment, and source properties are outside the matching set.

Covariate	References	Target roster	Matched targets
Observed z magnitude	15.71/17.11/21.08	13.29/19.40/20.00	15.67/17.07/19.97
Sérsic index	1.00/4.34/6.00	0.50/2.60/6.00	0.66/4.27/6.00
Axis ratio	0.15/0.70/0.95	0.052/0.61/1.00	0.20/0.70/0.93
r-band PSF FWHM (arcsec)	1.01/1.27/1.81	0.98/1.39/1.72	1.00/1.27/1.70
r-band depth (mag)	24.10/24.47/25.75	23.79/24.30/26.44	24.10/24.41/25.65

A13 $P_{\rm src}$ source construction and design

For the regularised COSMOS F814W source construction defined in Section 3.3 (Mandelbaum et al. 2012), we report the realised map-quality diagnostics and draw-count pilot below.

Of 81,499 catalogue rows, 770 satisfy the source criteria and 144 are selected without detector outcomes for the final experiment. The maximum structured–Sérsic centroid separation after construction is 7.34×10^{-5} arcsec, the minimum retained positive-flux fractions are 0.9010 and 0.9561 for the structured and Sérsic arms, respectively, and their source-map pixel correlations range from 0.6190 to 0.9985 (median 0.9767). These diagnostics characterise construction fidelity; they span visibly similar and dissimilar source-map pairs within the design.

A separate pilot used 1,584 pairs formed from 144 targets and 144

sources that do not appear in the final experiment. Its joint distribution over the nine target cells, eleven Einstein radii, and sixteen source strata is identical to the final design. The pilot and final samples are disjoint in pair, underlying-scene, target, spatial, physical, coordinate, pixel-hash, source, map-hash, and ancestry fields. The number of photon draws was chosen before the final experiment was scored.

The pilot scores 64 photon draws per pair. Table A13 shows the empirical MCSE at each candidate count. The selection rule chooses the smallest passing count, so eight draws are selected before final scoring.

Table A13. P_{src} design-pilot photon-draw selection. MCSE values and the target 0.01 are on the probability scale; 0.01 equals one percentage point. Eight is the smallest candidate count.

Draws	Measured MCSE	Pass
8	0.000683	Yes
16	0.000633	Yes
32	0.000387	Yes
64	0.000255	Yes

The final design contains 1,584 distinct targets in 1,219 spatial groups, with the 99 target-cell–radius blocks and balanced source reuse specified in Section 3.6. Section 4.8 defines the corresponding weighting and multiway uncertainty.

Before photon draws or detector scoring, all 1,584 paired expectation images pass the convergence comparison at oversampling factors 4, 8, and 16. The maximum consecutive relative L_1 difference is 0.002212, below the 0.01 criterion. The eight selected photon draws then produce 12,672 pair-draw rows per arm for the estimate reported in Section 5.6.

Under the reduced-cohort equal-cell rule, leave-one-source-out estimates range from +0.524813 to +0.580140 percentage points, and leave-one-spatial-group estimates range from +0.521104 to +0.576299 points. No deletion changes the sign. These are influence diagnostics rather than additional confidence intervals.

A14 Noiseless known-source transfer

We tested the preparation route with 16 noiseless known-source cases covering the retained source strata. Against an independently spatially refined analytic reference, the maximum normalized L_1 discrepancy is 0.08335 per cent and the maximum production centroid residual is 3.72×10^{-5} arcsec. All cases meet the global-L_1, centring, and normalization adequacy criteria, using limits of 1 per cent and 3×10^{-4} arcsec for the first two quantities. Under the stricter diagnostics, the local-pixel criterion is met in 1/16 cases and the shape criterion in 0/16. The global and stricter summaries therefore characterize different aspects of the same 16-case numerical check.

A15 Renderer reconciliation

The renderer test contains 142 scenes: the 43 matched-design injections, the order-48 scene, and one scene from each cell–radius stratum, with one overlap among these groups. This sample tests numerical implementation across the design but does not cover every production scene. Every scene passes replay and production quadrature convergence, independent optical-reference convergence after deterministic refinement where required, comparison with an effective post-pixel Gaussian reference, and comparison of production-support SIE deflections with an independent implementation. The effective-PSF comparison jointly changes operation order and FFT boundary handling and therefore tests their combined implementation.

The maximum and 95th-percentile production/reference relative L_1 differences are 0.007695 and 0.005802. The maximum centroid difference is 0.02293 native pixel, the minimum pixel correlation is 0.999952, and the maximum production-support SIE vector error is 9.06×10^{-9} arcsec. These bounds summarize numerical agreement for the audited scenes within the tested support.

The softened near-circular branch occurs in zero of the 49,500 realised G_{rec} scenes and zero of the 1,584 P_{src} pairs. Production-support SIE tests pass; the separate boundary-specific test does not meet its predefined criterion.